

Université Paris 1 Panthéon-Sorbonne

École d'Économie de la Sorbonne

Master 2 Development Economics

Import Competition and Markup Discipline

Evidence from Turkish Manufacturing

Master's Thesis

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Abstract

Chinese final-goods competition disciplines incumbent markups in exposed Turkish manufacturing sectors. I combine ORBIS/AMADEUS firm accounts with a shift-share instrument that uses 2009 Turkish exposure to Chinese imports and later Chinese export growth to other high-income markets. In the 2011–2019 baseline IV sample, a 10 percentage point increase in instrumented Chinese import penetration reduces annual log-markup growth among observed incumbent firms by 0.046 log points, with a coefficient of -0.462 ($SE = 0.226$). The effect is concentrated above the 80th percentile of the markup-adjustment distribution and is stronger in more concentrated sectors, consistent with residual-demand discipline. A causal-core indirect-inference exercise reproduces the main IV moments and leaves no residual output-IV signal after removing the fitted mechanism. By contrast, Chinese input exposure does not robustly lower within-firm markups; it mainly helps explain market-share reallocation among continuing firms. Rotemberg diagnostics show that the IV is driven by high-weight exposed cells, so the estimates speak to observed incumbents, not to the aggregate welfare effect of China's rise.

Keywords: Oligopoly, variable markups, shift-share instrument, import competition, firm dynamics, emerging markets

JEL: F14, F16, L11, L13, L22, O14

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1 Introduction

Chinese final-goods competition reduces markup growth among observed incumbent firms in exposed Turkish manufacturing sectors. In my baseline IV specification, a 10 percentage point increase in instrumented Chinese import penetration lowers annual log-markup growth by 0.046 log points, with a coefficient of -0.462 ($SE = 0.226$). The response is not uniform: it is concentrated above the 80th percentile of the markup-adjustment distribution and is stronger in more concentrated sectors. These patterns suggest that foreign competition compresses large positive markup adjustments in markets where residual-demand forces should be strongest.

Turkey is a useful setting for this question because Chinese exposure can work through two opposing channels. Since 2010, Turkish manufacturing has faced rising Chinese import penetration in sectors that are often concentrated and reliant on imported intermediates. Chinese final goods can tighten residual demand and reduce incumbents' pricing power.² Chinese intermediates can also lower marginal costs for firms that use imported inputs.³ Markups are the natural outcome for studying this tension because they connect trade shocks to prices, allocative efficiency, rent distribution, and industrial dynamics.

The empirical problem is that these channels, and the composition of the observed firm panel, are easy to confuse. Observed average markups may change because surviving firms adjust prices, because market shares move across continuing firms, or because firms disappear from ORBIS/AMADEUS. In this dataset, disappearance is not the same object as validated economic exit. A regression of markup changes on import penetration can therefore mix incumbent discipline with sample turnover and reallocation.

I separate these margins using four design choices. First, I instrument changes in Chinese import penetration with a shift-share IV that interacts 2009 Turkish sectoral exposure shares with Chinese export growth to other high-income markets. Second, I construct a separate input-supply shifter from predetermined Turkish input-output linkages, so the output-competition estimate is not asked to carry the input-cost channel. Third, I use a control-function diagnostic based on lagged EBIT margin to assess whether selective observed survival changes the within-incumbent result. Fourth, I connect the mean IV estimate to grouped IV quantiles and a sectoral decomposition into

²Tougher import competition increases the elasticity of residual demand and compresses markups (Edmond et al., 2015; Melitz & Ottaviano, 2008).

³Aghion et al. (2024) emphasizes these opposing firm-level responses to Chinese import exposure.

within-firm adjustment, reallocation, and observed entry and exit. Rotemberg-weight diagnostics show that the effective identifying variation is concentrated in a small set of highly exposed sector-year cells; the estimates should therefore be read as local evidence from those cells rather than as a manufacturing-wide average effect.

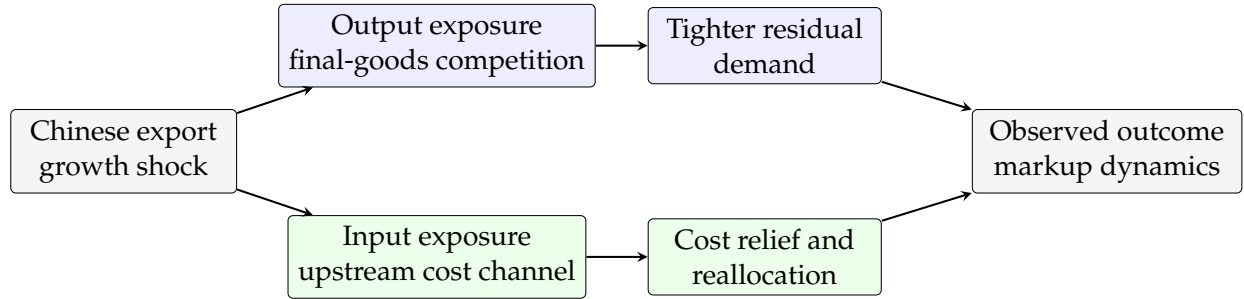


Figure 1: Conceptual map: output competition, input exposure, and markup dynamics

Notes: The empirical design treats output exposure as the excluded source of variation for Chinese import penetration and includes the input-supply shifter as a separate channel. The main coefficient measures the local response of markup growth among observed incumbent firms, while the decomposition and mechanism exercises study how input exposure and reallocation shape sector-level interpretation.

The results follow the map in Figure 1. Output-market Chinese import penetration reduces markup growth among observed incumbents. The selection diagnostic leaves the baseline coefficient close to unchanged, moving it from -0.462 to -0.454 , so observed survival does not appear to explain away the within-incumbent result. The grouped IV quantile estimates show little movement through the middle of the markup-adjustment distribution but large negative effects above the 80th percentile. At the sector level, total markup effects are weaker because reallocation partly offsets within-firm discipline. The input-supply shifter does not generate a robust within-firm markup response, but it helps explain market-share reallocation toward firms with stronger dependence on Chinese inputs.

The paper's central contribution is therefore narrow but useful: Chinese final-goods competition disciplines incumbent markups in exposed Turkish manufacturing sectors, while input exposure mainly shapes reallocation among continuing firms. This statement is deliberately local. The design identifies relative exposure effects across sector-year cells with high identifying weight; when the five most influential Rotemberg-weight sectors are excluded, the first stage weakens from about 24.8 to 3.2 and the second-stage estimate becomes unstable. The evidence therefore supports a local claim about influen-

tial exposed cells, not the common aggregate effect of China’s rise on all Turkish manufacturing or consumer welfare.

The closest papers are De Loecker et al. (2016) and Amiti and Heise (2025). De Loecker et al. (2016) show how trade liberalization affects prices, marginal costs, and markups using detailed Indian product data. I study annual markup dynamics among firms exposed to the China shock in Turkey, where accounting data require an explicit distinction between incumbent adjustment, observed sample survival, and sectoral reallocation. Amiti and Heise (2025) focus on import competition and market concentration in the United States. I instead ask whether imports discipline pricing power directly, whether this discipline is strongest where residual-demand forces should matter most, and whether upstream input exposure offsets or reallocates the output-market effect.

These contributions connect three literatures. Quantitative trade models with oligopolistic competition and variable markups emphasize that trade costs and foreign competition affect markups through residual demand, market shares, and pass-through (Atkeson & Burstein, 2008; de Blas & Russ, 2015; Gaubert & Itskhoki, 2021). The China-shock literature uses shift-share designs to estimate how rising Chinese competition affects local labor markets, employment, innovation, and firm outcomes (Acemoglu et al., 2016; Aghion et al., 2024; Autor et al., 2013, 2014; Bloom et al., 2016; Pierce & Schott, 2016). Work on industry dynamics and markup aggregation shows why separating within-firm adjustment from composition matters (Baqaei & Farhi, 2020; Burstein et al., 2025; Ericson & Pakes, 1995; Hopenhayn, 1992; Melitz, 2003). I add evidence on the dynamics behind these outcomes by showing how final-goods competition moves markups within observed survivors and how those movements are partly offset by reallocation.

The rest of the paper proceeds as follows. Section 2 describes Turkey’s industrial-development setting. Section 3 develops the oligopoly framework linking import competition, input exposure, and markups. Section 4 describes the Turkish firm panel, trade data, and markup measurement. Section 5 presents the IV design, selection diagnostic, decomposition, and grouped-quantile strategy. Section 6 reports the main incumbent and sector-level results. Section 7 evaluates threats to identification, robustness, and the indirect-inference validation exercise. Section 8 studies mechanisms and auxiliary outcomes. Section 9 concludes.

2 Turkey's Industrial Development Context

Turkey provides a useful setting for studying the pro-competitive effects of Chinese import competition. During the 2010s, Turkish manufacturing was exposed to rising competitive pressure from Chinese final-goods imports, while domestic production remained organized around a highly unequal firm-size distribution. This combination is important for the empirical design. If firms were atomistic, import competition would mainly operate through sectoral price indices and market selection. If instead a non-negligible part of domestic sales is accounted for by large firms, foreign competition can also discipline incumbent markups by reducing residual-demand power.

Figure 2 summarizes the evolution of Chinese import penetration across Turkish manufacturing sectors. The average sector was more exposed than the median sector, indicating that Chinese penetration was concentrated in a subset of sectors rather than evenly distributed across manufacturing. This dispersion is useful for identification: the empirical strategy exploits differential sectoral exposure to Chinese import growth, rather than a single aggregate China shock affecting all Turkish manufacturers symmetrically.

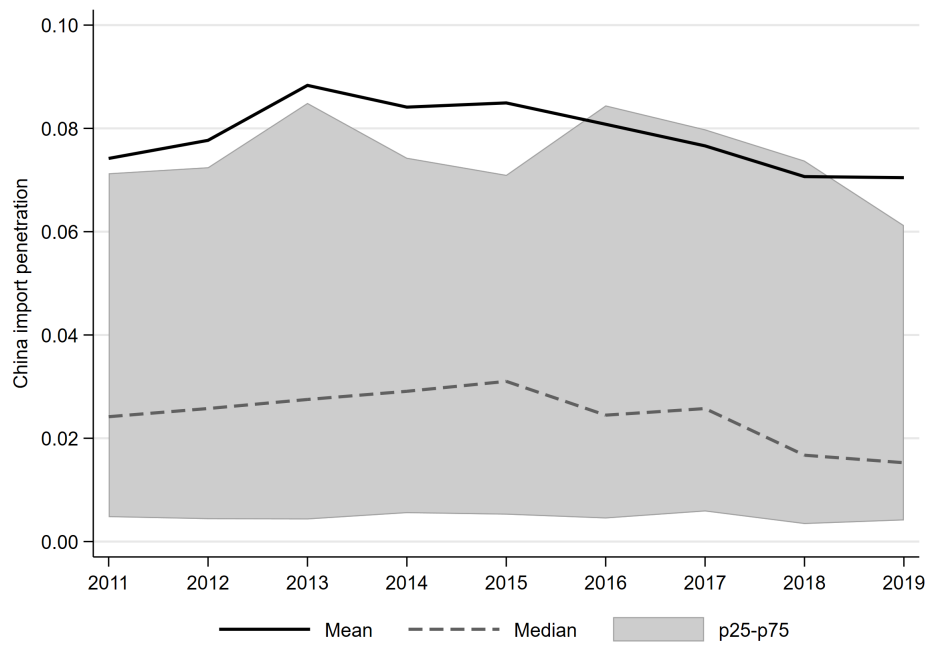


Figure 2: Chinese import penetration across Turkish manufacturing sectors

Notes: The figure reports the mean, median, and interquartile range of Chinese import penetration across ISIC4 manufacturing sectors.

A second feature of the setting is strong firm-size granularity. Figure 3 plots the Lorenz curve of domestic sales among Turkish manufacturing firms in 2011. The curve lies far below the equality line, with a Gini coefficient of about 0.84. This indicates that a small upper tail of firms accounts for a disproportionate share of domestic sales. The figure should not be interpreted as direct evidence of Cournot behavior, but it makes an atomistic benchmark unattractive: large firms are economically important enough that their pricing and markup responses are likely to matter for aggregate and sectoral outcomes.

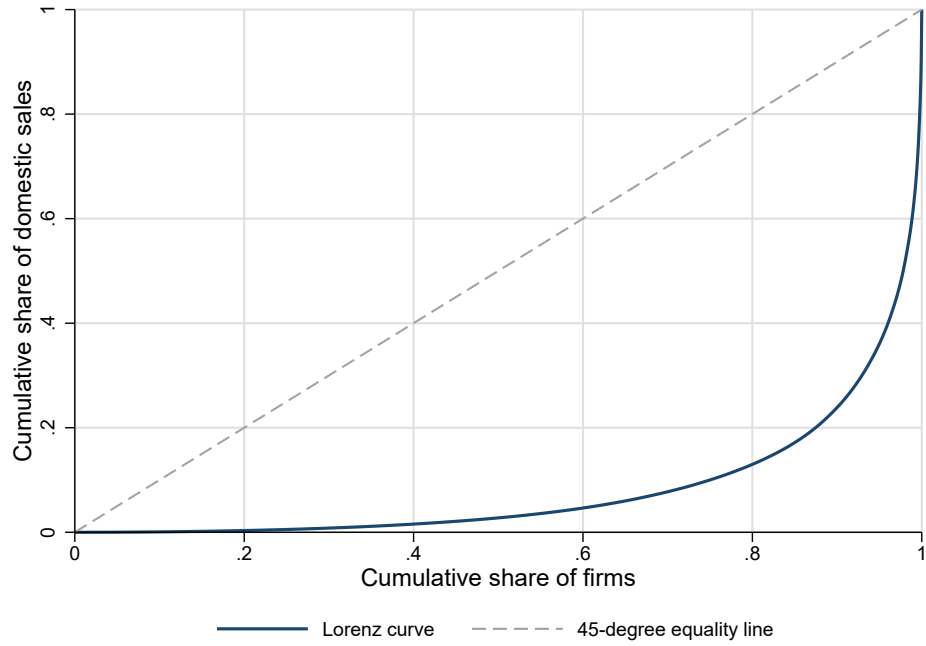


Figure 3: Lorenz curve of domestic sales among Turkish manufacturing firms

Notes: Firms are ranked from smallest to largest by domestic sales share within ISIC4-year cells. Gini = 0.836; N = 6,112. The dashed line is the 45-degree equality benchmark. The sample is restricted to firms with positive domestic sales in 2011.

The same point appears within sectors. Figure 4 reports CR4 and HHI statistics across ISIC4-year cells, restricting attention to cells with at least ten observed firms to avoid mechanically inflated concentration measures in thin cells. Even after this restriction, concentration is substantial: the mean CR4 is about 0.58, the median CR4 is about 0.57, and the 75th and 90th percentiles are about 0.73 and 0.86, respectively. HHI values are lower in level, as expected, but also indicate meaningful concentration in the upper tail of sector-year cells. Thus, the relevant empirical environment is not one of uniformly fragmented manufacturing sectors. Many sectors contain a small number of dominant firms alongside a competitive fringe.

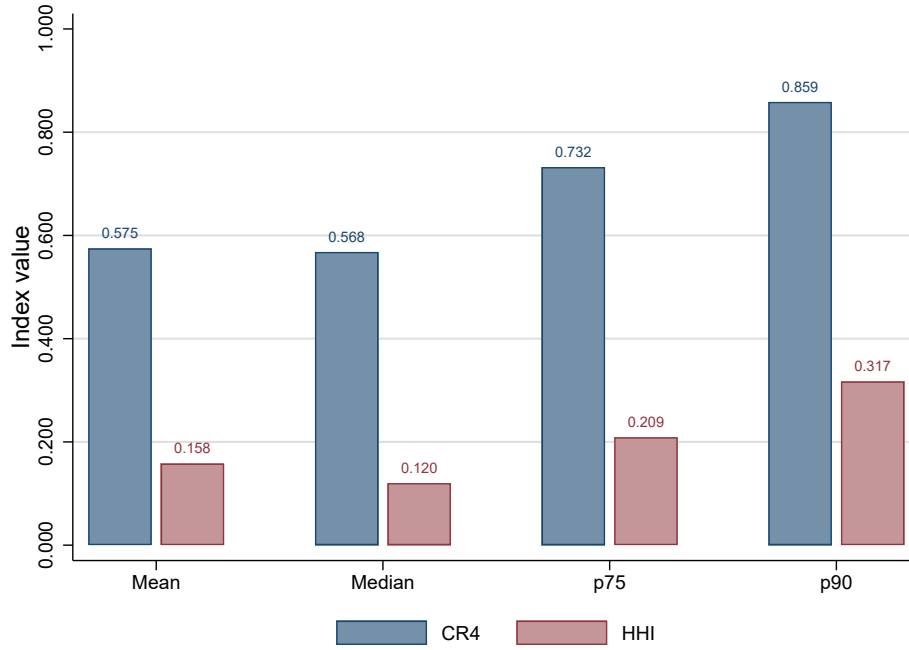


Figure 4: Sector-year concentration in Turkish manufacturing

Notes: The figure reports the mean, median, 75th percentile, and 90th percentile of CR4 and HHI across ISIC4-year cells. The sample is restricted to 2011–2019 and to sector-year cells with at least ten observed firms.

These facts motivate the Cournot oligopoly structure for Turkish manufacturing. The model below provides a tractable microfoundation for variable markups under finite-firm competition.

3 Theoretical framework

3.1 Model

This section presents the minimum theoretical backbone needed for the empirics. The model follows Atkeson and Burstein (2008) and Carvalho and Tahbaz-Salehi (2019), with domestic firms competing in quantities under nested CES demand and using imported intermediates in production. Detailed derivations of the nested CES system, the perceived elasticity formula, and the exit cutoff are in the Appendix A.

Environment and demand. The economy contains a continuum of sectors $j \in [0, 1]$.

In each sector, a finite set of domestic firms $i = 1, \dots, N_j$ produces differentiated varieties and competes under Cournot. A representative household allocates expenditure across sectoral composites with elasticity $\eta > 1$, so sectoral absorption is

$$D_j = \alpha_j \left(\frac{P_j}{P} \right)^{1-\eta} Y.$$

Within each sector, expenditure is split between a domestic composite and a foreign composite with elasticity $\nu_j > 1$:

$$Q_j = \left[(\Lambda_j^D)^{\frac{1}{\nu_j}} (Q_j^D)^{\frac{\nu_j-1}{\nu_j}} + (1 - \Lambda_j^D)^{\frac{1}{\nu_j}} (Q_j^F)^{\frac{\nu_j-1}{\nu_j}} \right]^{\frac{\nu_j}{\nu_j-1}}.$$

Within the domestic segment, varieties are aggregated with elasticity $\rho_j > 1$:

$$Q_j^D = \left(\sum_{i=1}^{N_j} \xi_i^{\frac{1}{\rho_j}} q_{ij}^{\frac{\rho_j-1}{\rho_j}} \right)^{\frac{\rho_j}{\rho_j-1}}.$$

I maintain $\rho_j > \nu_j > \eta$, so substitution is strongest across domestic varieties, weaker between domestic and foreign composites, and weakest across sectors. This ordering is what makes the output-competition effect stronger for firms with larger market shares and for more concentrated sectors.

Production and the input channel. Each firm produces according to

$$q_{ij} = \varphi_{ij} L_{ij}^{\beta_j} M_{ij}^{1-\beta_j}, \quad 0 < \beta_j < 1,$$

which implies marginal cost

$$\text{mc}_{ij} \propto \varphi_{ij}^{-1} w^{\beta_j} (P_j^M)^{1-\beta_j}.$$

Log changes in intermediate-input prices propagate through the Turkish input-output network, so a reduced-form approximation to marginal-cost growth is

$$d \ln \text{mc}_{ij} \approx -d \ln \varphi_{ij} + \beta_j d \ln w + (1 - \beta_j) \sum_k l_{jk} \omega_k^{CHN} d \ln P_k^{CHN}. \quad (1)$$

This expression is the basis for the empirical input-supply channel: Chinese input shocks lower costs more in sectors with stronger upstream exposure.

Oligopoly and markups. Let $\tilde{s}_{ij}$ denote firm i 's share within the domestic segment and s_{ij} its share in total sectoral expenditure. Under Cournot competition, the firm-specific perceived elasticity is

$$\sigma_{ij}^C = \left(\frac{1 - \tilde{s}_{ij}}{\rho_j} + \frac{\tilde{s}_{ij} - s_{ij}}{\nu_j} + \frac{s_{ij}}{\eta} \right)^{-1}.$$

The resulting optimal markup is $\mu_{ij} = \sigma_{ij}^C / (\sigma_{ij}^C - 1)$, or, more usefully for interpretation,

$$\frac{1}{\mu_{ij}} = \left(1 - \frac{1}{\rho_j} \right) - \left[\frac{1}{\Lambda_j^D} \left(\frac{1}{\nu_j} - \frac{1}{\rho_j} \right) + \left(\frac{1}{\eta} - \frac{1}{\nu_j} \right) \right] s_{ij}. \quad (2)$$

Aggregating across firms with the market-share-weighted harmonic mean implies

$$\frac{1}{\mu_j} = \left(1 - \frac{1}{\rho_j} \right) - \left[\frac{1}{\Lambda_j^D} \left(\frac{1}{\nu_j} - \frac{1}{\rho_j} \right) + \left(\frac{1}{\eta} - \frac{1}{\nu_j} \right) \right] \text{HHI}_j. \quad (3)$$

Sectoral markups therefore rise with concentration, and the slope of that relationship depends on the domestic expenditure share Λ_j^D . Stronger foreign competition lowers Λ_j^D , steepens residual demand, and disciplines markups most in sectors where concentration is already high.

Dynamic selection. Following Aguirregabiria et al. (2021), a firm that remains active in period t receives current profits and retains the option value of future operation. If the firm stays active, its continuation value satisfies

$$V(x_{ijt}) = \max \{0, \pi(x_{ijt}) + \beta \mathbb{E}[V(x_{ij,t+1}) \mid x_{ijt}]\}.$$

Trade shocks affect both current profitability and continuation values, so observed firm-level outcomes among survivors are selected on latent state variables. This is why the empirical strategy later treats the selection diagnostic as a reduced-form check for non-random sample survival rather than as a structural estimation of the exit problem.

3.2 Testable predictions and empirical mapping

The static model gives a useful map for the empirical design. The firm-level inverse-markup condition in equation (2) shows that markups depend on the domestic expenditure share and on the firm's market-share state, while the sectoral aggregation in equation (3) shows that the same residual-demand force is amplified by concentration. A rise in Chinese import penetration reduces the domestic share of spending and makes residual demand more elastic for Turkish incumbents. The first prediction is therefore a disciplining effect of output-market import competition on incumbent markup growth. Because equations (2)–(3) load this channel through firm shares and HHI_j , the effect should be stronger where initial market power is larger, especially in more concentrated sectors and among firms with market-share states associated with pricing power.

The input channel is different. The marginal-cost expression in equation (1) implies that cheaper or more abundant Chinese intermediates affect firms through production costs, with the strength of the channel depending on upstream input exposure. These cost changes can also reallocate sales across producers with different cost positions. The model therefore does not require the input channel to produce the same within-firm markup response as final-goods competition. I treat the input-supply shifter as a separate control in the baseline markup regressions and use the sectoral decomposition to study whether it operates mainly through reallocation.

The empirical specification translates these comparative statics into annual first differences. The main outcome is

$$\Delta \log \hat{\mu}_{ijt} = \log \hat{\mu}_{ijt} - \log \hat{\mu}_{ij,t-1}.$$

For small annual changes, this is the first-difference counterpart of equations (2) and (3).⁴ A predicted increase in inverse markups corresponds to lower log-markup growth. The coefficient on instrumented import-penetration growth is therefore interpreted as a local reduced-form response of incumbent markup growth to an output-competition shock, conditional on fixed effects, lagged controls, and the input-supply shifter. Because the instrument is a shift-share design, the identifying variation is relative across sector-year cells and is most informative for the cells receiving the largest Rotemberg weights.

⁴Appendix A.4 gives the formal bridge from the static inverse-markup condition to the first-difference estimating equation.

The grouped-IV quantile regressions and concentration interactions test whether discipline is concentrated where the model predicts residual-demand effects to be strongest. The sectoral decomposition separates within-firm adjustment from reallocation and observed sample turnover, so sector-level markup changes need not mirror the average incumbent response. Finally, because markup growth is observed only for continuing reporters, the selection diagnostic in Section 5.2 is interpreted as a reduced-form check for non-random survival.

4 Data and Measurement

In this section, I describe the data sources, define the samples and key variables I use in the analysis, and present summary statistics.

4.1 Data sources

Firm-level data. The ORBIS/AMADEUS dataset contains harmonized balance-sheet, income-statement, and profit-and-loss information for firms. I use the Turkish manufacturing panel from 2009 to 2023 and map firms from NACE Rev.2 into 4-digit ISIC sectors using a concordance. The raw manufacturing panel contains 15,957 firms and 78,921 firm-year observations. The baseline IV estimates use 2011–2019, because the sectoral trade and output variables needed to construct import penetration, apparent consumption, and the shift-share instrument are consistently available over that window. The 2009–2010 firm observations provide lags and the 2009 exposure shares; post-2019 firm observations are retained for descriptive coverage and sample-turnover context but do not enter the baseline trade-IV estimates. Because the main dependent variable is markup growth, the estimation sample is necessarily smaller than the raw panel: a firm-year enters the baseline markup regressions only when current and lagged accounting variables are both observed and revenue, COGS, and capital are positive. After applying these restrictions, winsorizing extreme variables at the 1st and 99th percentiles, and adding lagged controls, the main firm-level specifications use about 31,000 firm-year observations.

The panel is unbalanced, and I do not impute intermittent reporting gaps. I define *observed exit* as disappearance from ORBIS/AMADEUS between $t - 1$ and t . This operational definition is useful for studying sample composition, but it is not equivalent

to validated legal or economic firm death. Throughout the thesis, results on entry and exit should therefore be read as evidence on observed sample turnover unless otherwise stated.

A relevant institutional break occurs in 2016, when Turkey implemented a large one-time increase in the statutory minimum wage. The shock mattered especially for labor-intensive formal firms because it raised reported payroll costs and may have changed the incentives to remain in formal reporting systems. In ORBIS/AMADEUS, this can appear as a coverage change or reporting attrition even when it is not validated firm death. To avoid attributing such differential post-2016 coverage changes to Chinese import competition, the baseline specifications control for pre-2016 sectoral labor intensity interacted with a post-2016 indicator. The control is predetermined with respect to the post-2016 episode and absorbs the fact that labor-intensive sectors were more exposed to this institutional shock.

Trade data. My main treatment variable is Chinese import penetration at the 4-digit ISIC sector-year level, defined as

$$\text{China IP}_{jt} = \frac{\text{Import}_{jt}^{\text{CHN} \rightarrow \text{TR}}}{\text{Output}_{jt} + \text{Import}_{jt}^{\text{World} \rightarrow \text{TR}} - \text{Export}_{jt}^{\text{TR} \rightarrow \text{World}}} \quad (4)$$

where the denominator corresponds to domestic absorption. I construct the numerator from CEPII BACI HS96 bilateral trade data at the HS 6-digit level and map products to 4-digit ISIC sectors using the HS–ISIC concordance generated with the `concordance` package of Liao et al. (2020). Appendix B.1.2 documents the mapping procedure and the treatment of many-to-many product-industry links. Sectoral output and trade aggregates used in the denominator come from UNIDO ISDB4/INDSTAT4. The treatment in the regressions is the annual change in this import-penetration rate.

For the input-supply channel, I propagate the output-side shift-share shock through Turkish input-output linkages. The weights come from the 2003 Turkish input-output table in WIOD and are kept fixed over the sample period. Using pre-period weights is a deliberate trade-off: it cannot capture subsequent changes in sourcing networks, but it preserves a predetermined exposure matrix and avoids loading the upstream shifter with contemporaneous endogenous restructuring.

Other data. I use several TURKSTAT datasets to construct controls and deflators. The 3-digit Producer Price Index (PPI) is used to deflate nominal variables. Personnel-

cost and value-added data from the Annual Industry and Service Statistics are used to construct pre-period sectoral labor intensity. Across the empirical sections, the firm-level outcome data are therefore combined with a sector-year treatment design and a smaller set of effective identifying cells.

Table 1 reports summary statistics for firm-level and sector-level variables used in the baseline and mechanism analyses.

Table 1: Summary Statistics

Variable	N	Mean	SD	P50	Min	Max
Panel A: Firm-level						
$\Delta \ln \mu$	31339	0.002	0.141	0.001	-3.873	4.184
Observed exit	31339	0.083	0.276	0.000	0.000	1.000
EBIT margin	31339	6.890	8.818	5.670	-95.350	95.750
$\ln(\text{Size})$	31339	15.181	1.632	15.132	7.399	22.085
Leverage	31339	0.229	0.205	0.193	-0.053	2.024
Liquidity ratio	31339	1.411	2.714	0.910	0.000	99.470
Age	31339	5.772	3.317	5.000	2.000	15.000
Exporter	31339	0.697	0.460	1.000	0.000	1.000
Panel B: Sector-level						
ΔIP	609	-0.000	0.027	-0.000	-0.224	0.213
Output IV	609	-0.001	0.008	-0.000	-0.071	0.066
Z^{Input}	609	-0.000	0.003	-0.000	-0.009	0.011
Labor share (pre)	609	0.510	0.092	0.510	0.260	0.887
HHI (domestic)	609	0.200	0.220	0.114	0.001	1.000

Panel A reports firm-level variables; Panel B reports sector-level variables.

4.2 Chinese import competition

Estimating the effect of changes in Chinese import penetration on firm outcomes by OLS is problematic for at least three reasons. First, Turkish sectoral demand or productivity shocks may affect both imports from China and domestic firm outcomes. Second, persistent sectoral characteristics may jointly shape exposure to Chinese competition and firm performance. Third, the treatment varies at the sector-year level while the main outcomes are observed at the firm-year level, so naive firm-level regressions risk over-

stating the amount of independent identifying variation.

To address these issues, I implement a shift-share instrumental-variable design following Autor et al. (2014) and the shock-level interpretation in Borusyak et al. (2022). The instrument combines exogenous shifts in Chinese export performance in other high-income markets with predetermined Turkish exposure shares. In the baseline, the exposure shares are fixed at $t_0 = 2009$. This construction is meant to isolate variation in Turkish import competition that comes from common China-driven shocks rather than from contemporaneous Turkish sectoral conditions.

Formally, let $M_{jt}^{\text{CHN} \rightarrow \text{HI}}$ denote Chinese exports to high-income countries in sector j at time t , and let $M_{jt}^{\text{World} \rightarrow \text{HI}}$ denote total world exports to high-income countries in the same sector and period. The shift component of the instrument is defined as the change in China's export share in high-income destination markets:

$$g_{jt} = \Delta \ln \text{Import Share}_{jt}^{\text{CHN} \rightarrow \text{HI}} = \Delta \ln M_{jt}^{\text{CHN} \rightarrow \text{HI}} - \Delta \ln M_{jt}^{\text{World} \rightarrow \text{HI}}$$

The share component is the baseline level of Chinese import penetration in Turkey in period $t_0 = 2009$. The shift-share instrument for changes in Chinese import penetration is therefore

$$Z_{jt}^{\text{Output}} = s_{j,\text{CHN}}^{2009} \times g_{jt}.$$

To capture the input-supply channel separately, I construct an additional control variable by weighting the sector-level output instrument by Turkish input-output linkages across sectors, following Amiti and Konings (2007):

$$Z_{jt}^{\text{Input}} = \sum_k w_{jk}^{t_0} \times Z_{kt}^{\text{Output}}$$

where $w_{jk}^{t_0}$ denotes the technical coefficient derived from the 2003 Turkish input-output table, defined as the share of sector k in total intermediate inputs used by sector j :

$$w_{jk}^{t_0} = \frac{\text{Domestic Use}_{jk} + \text{Imported Use}_{jk}}{\text{Total Intermediate Inputs}_j}$$

The relevant treatment cell in all IV specifications is therefore the sector-year, not the firm-year. In the baseline sample, the firm-level regressions reuse 609 effective sector-year cells spanning 95 ISIC4 manufacturing sectors. Sector fixed effects absorb time-

invariant sector heterogeneity, year fixed effects absorb aggregate shocks common across sectors, and placebo-lead tests assess whether future China shocks predict current outcomes. The main remaining threat is a sector-specific global demand or technology shock that simultaneously raises Chinese exports to other high-income markets and changes Turkish markups. The identifying assumption is that, conditional on fixed effects and controls, these residual sector-year shocks are orthogonal to the constructed shift. Because Rotemberg-weight diagnostics show that some sectors are much more influential than others, I interpret the IV estimates as local to the more exposed sector-year cells rather than as universal effects for all Turkish manufacturing sectors.

4.3 Markup estimation and other variables

Markups. I estimate markups using the production approach pioneered by Hall (1988) and developed by De Loecker and Warzynski (2012) and De Loecker et al. (2020). In the baseline implementation, I estimate a Cobb-Douglas revenue production function within ISIC4 sectors, use COGS as the flexible input, and recover the output elasticity of COGS from that production function. The markup estimator is

$$\mu_{it} = \theta_{ivt} \frac{P_{it}Q_{it}}{P_{vit}V_{it}},$$

where θ_{ivt} is the output elasticity of the flexible input, $P_{it}Q_{it}$ is firm revenue, and $P_{vit}V_{it}$ is expenditure on input v . COGS is the most consistently observed short-run variable input in ORBIS/AMADEUS and is therefore the baseline flexible input. Appendix B.1.1 reports the implementation details, a translog sensitivity check, and robustness exercises from Kirov et al. (2026).

Because accounting-based markup levels are sensitive to production-function assumptions, reporting quality, and the choice of flexible input, I focus on within-firm changes rather than levels (De Ridder et al., 2026). The main dependent variable is

$$\Delta \log \hat{\mu}_{it} \equiv \log \hat{\mu}_{it} - \log \hat{\mu}_{i,t-1}$$

This choice is meant to limit the influence of time-invariant reporting differences and level mismeasurement.

Labor Intensity. Labor intensity is constructed as the ratio of personnel costs to

value added factor cost, averaged over 2009–2015 at the 4-digit sector level. This time-invariant measure captures differences in production technology across industries. Following the institutional discussion above, the interaction of labor intensity with a post-2016 dummy captures differential exposure to the minimum-wage episode and the associated risk of formal-firm reporting attrition in ORBIS/AMADEUS.

Sectoral Markup. Following Burstein et al. (2025), sectoral markup is defined as the market-share-weighted harmonic mean of firm markups:

$$\mu_{jt} = \left(\sum_{i \in j} s_{ijt} \mu_{it}^{-1} \right)^{-1}$$

where $s_{ijt} = \frac{D_{it}}{\sum_{i' \in j} D_{i't}}$ is firm market share, which is firm domestic sales deflated by industry PPI. The harmonic mean is appropriate because it gives more weight to low-markup firms, which are more likely to be affected by import competition and selection. The change in sectoral markup is then defined as the first difference of the log of the sectoral markup.

5 Empirical strategy

The empirical strategy has four components. Section 4.1 states the baseline shift-share IV specification and the identifying assumptions, with particular emphasis on the fact that the effective variation comes from a limited set of sector-year cells. Section 4.2 asks whether selective survival in the observed panel materially changes the within-incumbent estimate. Sections 4.3 and 4.4 then connect the mean result to sectoral aggregation and distributional heterogeneity through a decomposition and grouped IV quantile regressions.

5.1 Identification strategy and baseline IV regression

My baseline IV specification estimates the effect of changes in Chinese import penetration on firm-level markup adjustments, controlling for the input-supply channel and

other covariates. The regression is specified as follows:

$$Y_{ijt} = \beta_O \Delta IP_{jt} + \beta_I Z_{jt}^{\text{Input}} + \gamma_1 \text{LaborShare}_{jt_0} \times \text{Post2016}_t + \gamma_2 \text{LaborShare}_{jt_0} + X'_{ijt-1} \gamma + \tau_t + \delta_j + \varepsilon_{ijt} \quad (5)$$

where Y_{ijt} is the outcome of interest (observed exit probability or markup adjustments), ΔIP_{jt} is the change in Chinese import penetration, Z_{jt}^{Input} is the control variable for the input-supply channel, LaborShare_{jt_0} is the pre-2016 averaged labor intensity, X_{ijt-1} are lagged firm-level controls, and τ_t and δ_j are year and sector fixed effects. The coefficient of interest is β_O , which measures the effect of output-market import competition after controlling for upstream input shocks and observables.

Identification comes from the output-side shift-share IV Z_{jt}^{Output} for ΔIP_{jt} , while Z_{jt}^{Input} remains included as a control. Two clarifications are important. First, the treatment cell is the sector-year and the baseline exposure share is fixed in 2009, so the firm-level regression does not create additional independent shocks beyond those sector-year cells. Second, the coefficient should be interpreted as local to the sector-year cells that receive the most identifying weight. Figure 5 shows that the largest Rotemberg-weight sectors are economically small but highly exposed to Chinese imports in 2009. This pattern disciplines the external-validity claim of the baseline estimate: the IV is informative about exposed cells that carry identifying variation, not about every manufacturing sector equally.

This limitation is substantive rather than cosmetic. The appendix exclusion exercise shows that once the most influential Rotemberg-weight sectors are removed, the first stage weakens sharply and the second-stage estimate becomes unstable in both sign and magnitude. I therefore treat the baseline IV as evidence that Chinese import competition disciplines markups in influential exposed sectors, while avoiding the stronger claim that the same estimate is stable across all plausible reallocations of identifying weight.

I also use two auxiliary identification checks to address sector-specific confounding. First, demand-control specifications add changes in high-income global demand and Turkish export demand for the sector. These controls ask whether the baseline result is really picking up foreign demand or export-market shocks rather than Chinese import competition. Appendix Table 13 shows that the import-penetration coefficient remains essentially unchanged. Second, pre-trend balance tests regress pre-period sectoral outcomes on future China shocks. Appendix Table 14 shows no meaningful predictive rela-

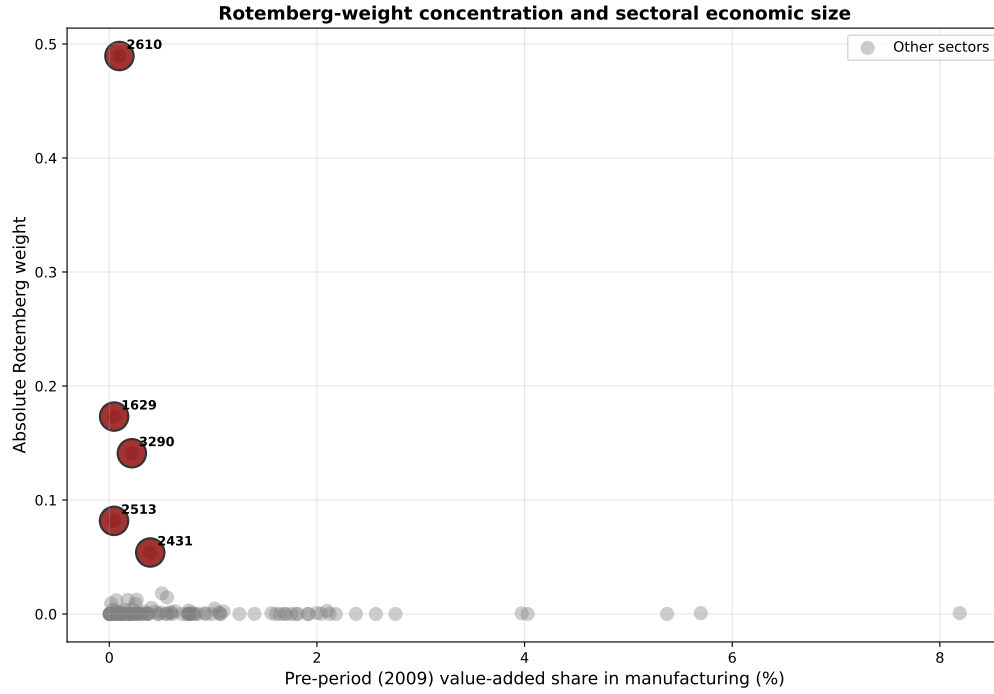


Figure 5: Top Rotemberg-weight sectors and pre-period economic size

tionship for the future output-side shock, which complements the lead-placebo evidence below.

The control for pre-2016 labor share interacted with the post-2016 dummy captures differential exposure to the minimum-wage episode, which may have affected the formal-firm coverage (Akcigit et al., 2020). Because firms with similar exposure structures may have correlated residuals, inference is clustered at the ISIC4 level and complemented, where available, with wild-cluster bootstrap p-values in the robustness section.

Table 2 summarizes the identifying claim and the main checks. The key point is that the design is credible for a local estimand: observed incumbent firms in sector-year cells where the shift-share instrument creates meaningful variation. The table also makes clear what the design does not do. It does not identify the aggregate effect of China's rise, and it does not turn ORBIS/AMADEUS disappearance into validated economic exit.

To assess the plausibility of the exclusion restriction, I conduct lead-placebo tests. If the instrument is valid, future realizations of the China shock should not predict current outcomes once fixed effects and controls are included. Conversely, significant placebo

Table 2: Identification summary

Component	What identifies the estimate	Evidence and interpretation
Excluded variation	2009 Turkish sector exposure to China interacted with Chinese export growth to other high-income markets.	The first stage is positive and precise: 0.892 (SE = 0.179), with a Kleibergen–Paap rk Wald F -statistic of about 24.8 (Table 4).
Main identifying assumption	Conditional on sector and year fixed effects and controls, the shift is orthogonal to residual Turkish sector-year shocks that also move markups.	Lead-placebo estimates are statistically indistinguishable from zero (Figure 6), reducing concern that future China shocks predict current markup changes.
Output versus input channel	The output-side shift-share variable instruments final-goods import penetration; the input-supply shifter enters separately as a control.	The input-supply shifter is weakly correlated with the output IV and does not drive the first stage. Residualizing the output IV on the input shifter leaves the markup coefficient unchanged in the appendix.
Effective variation	Treatment varies at the sector-year level, not the firm-year level.	The baseline uses 609 sector-year cells and 95 sectors. Rotemberg diagnostics show concentration of identifying weight, so the estimate is interpreted locally rather than as a uniform manufacturing-wide effect.
Selection in observed firm panel	Markup growth is observed only for firms that report in adjacent years.	Adding the EBIT-margin control function moves the coefficient from -0.462 to -0.454 (Table 5), so the estimate is not materially changed by this selection proxy.
Robustness of magnitude	The coefficient should survive reasonable changes in specification and outcome construction.	The sign and broad magnitude survive control-set changes, trimming, alternative markup estimation, and residualized-IV checks; long-lived-firm restrictions attenuate the estimate, consistent with removing part of the composition margin (Tables 11, 21, and 22).

coefficients would suggest pre-trends or omitted sectoral shocks that threaten the design.

Formally, for horizon $k \in \{1, 2, 3\}$, I estimate:

$$Y_{ijt} = \beta_O^k Z_{j,t+k}^{\text{Output}} + \beta_{I0} Z_{jt}^{\text{Input}} + \beta_{I1} Z_{j,t-1}^{\text{Input}} + \gamma_1 \text{LaborShare}_{jt_0} \times \text{Post2016}_t \\ + \gamma_2 \text{LaborShare}_{jt_0} + X'_{ijt-1} \gamma + \tau_t + \delta_j + \varepsilon_{ijt} \quad (6)$$

Under the identifying assumption, the placebo coefficients β_O^k should be statistically indistinguishable from zero, providing evidence against pre-trends and supporting the validity of the instrument. I summarize these placebo coefficients graphically using a coefficient plot with confidence intervals across horizons.

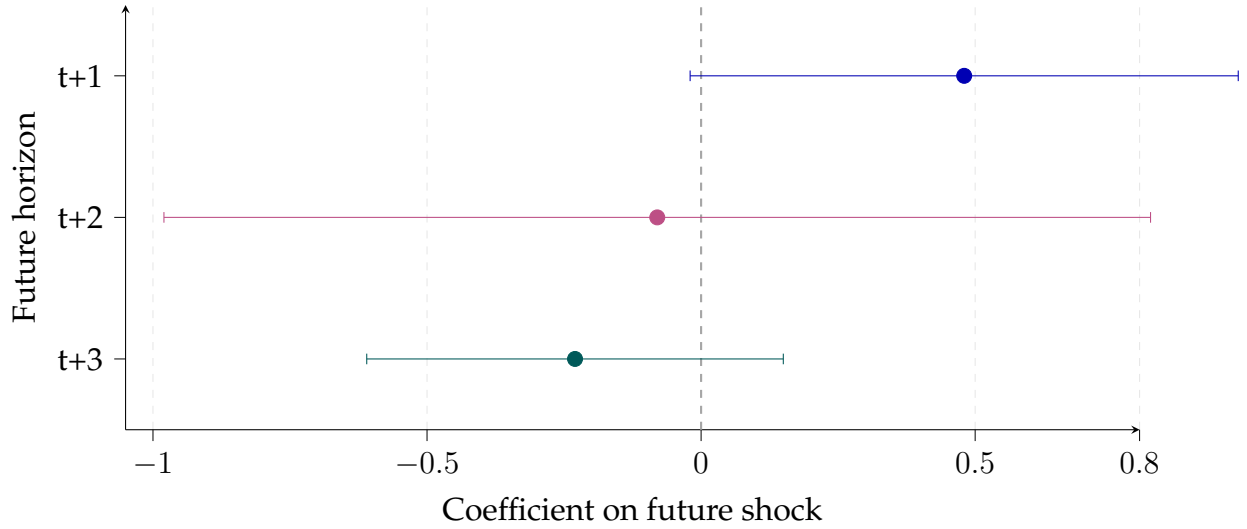


Figure 6: Placebo: future shocks predicting current markup changes

The results in Figure 6 support the validity of the design. Across the one to three-year horizons, the estimated placebo coefficients are small, not systematically signed, and statistically insignificant, suggesting no evidence of pre-trends or reverse causality. This provides credibility to the interpretation of the main IV estimates as capturing the causal effect of Chinese import penetration on firm outcomes.

5.2 Selection diagnostic

A key concern in the baseline regression is endogenous sample selection. Firm-level markup adjustments are observed only for firms that remain in the panel and continue

reporting. In ORBIS/AMADEUS, disappearance may reflect true economic exit, reporting gaps, or coverage changes. The empirical problem is the same in each case: if Chinese import competition affects observed survival and that margin is correlated with unobserved determinants of markup adjustment, then the baseline IV estimates mix a within-firm response with compositional change in the observed sample.

To address this issue, I adopt a semiparametric control-function diagnostic inspired by the selection approach in Backus (2020) and Pavcnik (2002). The objective is to test whether a flexible proxy for latent firm state materially changes the markup estimate: if observed survival depends on latent firm state and that state also affects markup adjustment, then the regression error among observed firms is not mean-independent of trade exposure.

I proxy the latent state with lagged EBIT margin, denoted $s_{i,t-1}$. This variable is observable, persistent, predictive of observed exit, and informative about subsequent markup adjustment, so it provides a practical control-function index for selection. Let the second-stage error decompose as $\varepsilon_{i(j)t} = \lambda_{jt} + \eta_{it}$, where λ_{jt} captures sector-year shocks and η_{it} is an idiosyncratic component. Among surviving firms, I model the selection-induced component as a flexible function of the lagged state proxy,

$$\mathbb{E}[\eta_{it} \mid s_{it-1}] = g(s_{it-1}),$$

and approximate $g(\cdot)$ with a third-order polynomial in s_{it-1} . The cubic is flexible enough to capture nonlinear selection on profitability without overfitting a relatively small support of the state proxy. The diagnostic second-stage regression therefore includes this control function:

$$\begin{aligned} \Delta \ln \mu_{i(j)t} = & \beta_O \widehat{\Delta IP}_{jt} + \beta_I Z_{jt}^{\text{Input}} + \gamma_1 \text{LaborShare}_{jt_0} \times \text{Post2016}_t \\ & + \gamma_2 \text{LaborShare}_{jt_0} + X'_{ijt-1} \gamma + \tau_t + \delta_j + g(s_{it-1}) + \varepsilon_{i(j)t} \end{aligned} \quad (7)$$

Conditional on the flexible control function $g(s_{it-1})$, the remaining variation in the instrumented import penetration shock gives a reduced-form check of whether the baseline estimate is sensitive to selection on the observed state proxy. Thus, β_O should still be interpreted as a within-incumbent effect among observed survivors, with the control function showing that this estimate is not materially changed by a practical selection proxy rather than proving that all selection bias has been removed.

Table 3: Validation of Lagged EBIT Margin as State Proxy

	(1) Persistence	(2) Observed Exit (linear)	(3) Observed Exit (quadratic)	(4) Markup Adjustments	(5) Markov Test
EBIT Margin _{<i>t</i>-1}	0.0437*** (0.0164)	-0.000453** (0.000229)	-0.000524** (0.000237)	-0.00876*** (0.000528)	-0.000977*** (0.000306)
EBIT Margin _{<i>t</i>-1} ²			-0.0000133*** (0.00000482)		
EBIT Margin _{<i>t</i>-2}					0.0000519 (0.000287)
Observations	41,113	41,150	41,150	41,150	29,326
<i>R</i> ²	0.649	0.341	0.341	0.356	0.367

Standard errors clustered at firm level in parentheses. All specifications include firm, sector (ISIC4), and year fixed effects. Controls: leverage, liquidity ratio, log firm size, age, export status, and labor share $\times$ Post-2016 indicator; sample 2011–2019.

Column (1) tests persistence; columns (2)–(3) test predictive power for observed exit; column (4) tests markup adjustments among survivors; column (5) reports the Markov test (lagged EBIT margin at $t - 2$, p-value 0.857).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The validity of lagged EBIT margin as a state proxy is examined in Table 3. The evidence supports its use. First, lagged EBIT margin is persistent over time, consistent with a slowly evolving firm state. Second, it predicts observed exit, as required for the diagnostic. Third, it is associated with future markup adjustments among survivors, indicating that it captures economically relevant heterogeneity in firms' dynamic responses. Finally, the Markov restriction is supported by the data: once lagged EBIT margin at $t - 1$ is included, the additional lag at $t - 2$ is not significant. This does not prove that EBIT margin is a sufficient statistic, but it supports its use as a practical reduced-form proxy for latent profitability.

5.3 Decomposition of sectoral markup changes

Firm-level regressions identify whether Chinese import penetration disciplines incumbents' pricing behavior, but they do not by themselves show how these responses aggregate to the sector level. This matters because sectoral markups may move differently from firm-level markups when import shocks also reallocate market shares across incumbents or change which firms are observed in adjacent years. To connect micro responses to sector-wide outcomes, I therefore decompose annual changes in sectoral markups into within-firm, reallocation, observed entry, and observed exit components. This decomposition is suited to environments with variable markups, where sectoral

markup movements depend jointly on firm-specific markup adjustment and changes in market shares.

Let S_t denote the set of firms observed in sector j at time t , $C = S_{t-1} \cap S_t$ the set of continuing firms, $E = S_t \setminus S_{t-1}$ observed entrants, and $X = S_{t-1} \setminus S_t$ observed exiters. Adapted from Melitz and Polanec (2015), the change in sectoral inverse markup can then be written as:

$$\Delta \mu_{jt}^{-1} = \underbrace{\sum_{i \in C} \bar{s}_{ij} \Delta \mu_{ij}^{-1}}_{\text{Within (continuers)}} + \underbrace{\sum_{i \in C} \Delta s_{ij} \bar{\mu}_{ij}^{-1}}_{\text{Between (continuers)}} + \underbrace{\sum_{i \in E} s_{ijt} \mu_{ijt}^{-1}}_{\text{Observed entry}} - \underbrace{\sum_{i \in X} s_{ij,t-1} \mu_{ij,t-1}^{-1}}_{\text{Observed exit}}$$

where $\bar{s}_{ij} = \frac{1}{2}(s_{ijt} + s_{ij,t-1})$ and $\bar{\mu}_{ij}^{-1} = \frac{1}{2}(\mu_{ijt}^{-1} + \mu_{ij,t-1}^{-1})$ are the average market share and inverse markup for firm i over the two periods. This decomposition allows me to isolate the contributions of different mechanisms to sectoral markup changes.

The within term captures markup adjustment among continuing firms, holding their market shares fixed at midpoint weights. It therefore measures the contribution of incumbent pricing responses to aggregate markup discipline. The reallocation term captures changes in sectoral inverse markups due to shifts in market shares across continuing firms, holding their markups fixed at midpoint values. This term is positive when market shares move toward firms with lower markups and negative when reallocation favors firms with higher markups. The observed-entry term measures the contribution of firms newly appearing in year t , while the observed-exit term captures the contribution of firms present in $t - 1$ that disappear in t . Because the decomposition is written in inverse markups, a positive component corresponds to a decline in sectoral markups, that is, a pro-competitive effect, while a negative component corresponds to an increase in sectoral markups, that is, an anti-competitive effect.

I compute these four components for each sector-year and then use them separately as dependent variables in the sector-level IV specifications. This allows the import-competition shock to affect aggregate markups through distinct channels. Formally, for each outcome $m \in \{\text{within-firm, reallocation, observed entry, observed exit}\}$, I estimate:

$$Y_{jt}^m = \beta_O^m \widehat{\Delta IP}_{jt} + \beta_I^m Z_{jt}^{\text{Input}} + \Gamma' W_{jt} + \delta_j + \tau_t + \epsilon_{jt}^m \quad (8)$$

where Y_{jt}^m is the sector-level outcome corresponding to component m , $\widehat{\Delta IP}_{jt}$ is the change in Chinese import penetration instrumented by the output-competition shock, Z_{jt}^{Input} is

the control for input supply channel, W_{jt} are sector-level controls, and δ_j and τ_t are sector and year fixed effects. The coefficient β_O^m captures the effect of import competition on the specific channel m .

This decomposition is central because the aggregate response of sectoral markups need not mirror the average firm-level response. Even if import competition disciplines continuing firms, the sectoral effect may be attenuated or reversed when market shares are reallocated toward dominant incumbents, or when observed entry and exit contribute in offsetting directions. Estimating the four terms separately therefore makes it possible to distinguish whether sectoral markup changes are primarily driven by incumbent adjustments, reallocation of market shares, or observed sample turnover. In addition, to assess the relative importance of each margin, I express each decomposition component as a share of the total absolute change in inverse sector markup.

5.4 Distributional effects and concentration heterogeneity

To move beyond average effects, I examine whether the impact of import competition varies across the distribution of firm-level markup adjustments. This is important because mean IV estimates may mask whether trade discipline is concentrated in the upper tail of markup-growth realizations, where firms are attempting the largest positive adjustments. I therefore complement the baseline specification with a grouped IV quantile approach Chetverikov et al. (2016) adapted to the settings in which the endogenous treatment varies at the sector-year level.

I define $\rho_{jt}^{(k)}$ as the k -th decile of the distribution of the within-sector-year distribution of firm-level markup adjustments among survivors. For each decile k , the regression is specified as:

$$\begin{aligned} \rho_{jt}^{(k)} = & \beta_O^{(k)} \widehat{\Delta IP}_{jt} + \beta_I^{(k)} Z_{jt}^{\text{Input}} + \gamma_1^{(k)} \text{LaborShare}_{jt_0} \times \text{Post2016}_t \\ & + \gamma_2^{(k)} \text{LaborShare}_{jt_0} + X'_{jt-1} \gamma^{(k)} + \tau_t^{(k)} + \delta_j^{(k)} + \nu_{jt}^{(k)} \end{aligned} \quad (9)$$

The grouped formulation is preferable to a standard firm-level IV quantile regression because the treatment is common within sector-year cells. Pooling firms directly would require rank-similarity assumptions across sectors that are difficult to justify in the presence of sector-level unobservables. By using sector-year quantiles as the dependent variable, the specification instead isolates how import competition shifts different

parts of the outcome distribution across groups.

Identification relies on the same shift-share design as in the baseline analysis. Sectoral changes in Chinese import penetration are instrumented using exogenous growth in Chinese exports to high-income destinations interacted with predetermined Turkish exposure shares, while controlling for the input supply channel. Conditional on fixed effects and controls, $\widehat{\Delta IP_{jt}}$ is therefore interpreted as exogenous variation in output-market competitive pressure.

Because sample quantiles are mechanically sensitive to the number of firms in each sector-year cell, particularly in smaller sectors, I correct for order-statistic bias (Neyman and Scott, 1948) by augmenting the specification with a flexible control function of the number of firm observations n_{jt} :

$$\begin{aligned} \rho_{jt}^{(k)} = & \beta_O^{(k)} \widehat{\Delta IP_{jt}} + \beta_I^{(k)} Z_{jt}^{\text{Input}} + \gamma_1^{(k)} \text{LaborShare}_{jt_0} \times \text{Post2016}_t \\ & + \gamma_2^{(k)} \text{LaborShare}_{jt_0} + X'_{jt-1} \gamma^{(k)} + \tau_t^{(k)} + \delta_j^{(k)} + g^{(k)}(n_{jt}) + \nu_{jt}^{(k)} \end{aligned} \quad (10)$$

where $g^{(k)}(\cdot)$ is approximated with a third-order polynomial in n_{jt} . This follows the correction proposed in Backus (2020), where order statistics can be spuriously correlated with group size when the number of firms is small relative to the number of quantiles.

I then study heterogeneity in the pro-competitive effect by interacting predicted import penetration with lagged domestic concentration measures such as HHI, CR10, and CR4:

$$\begin{aligned} \rho_{jt}^{(k)} = & \beta_{O1}^{(k)} \widehat{\Delta IP_{jt}} + \beta_{O2}^{(k)} \widehat{\Delta IP_{jt}} \times \text{Concentration}_{jt-1} + \beta_I^{(k)} Z_{jt}^{\text{Input}} + \gamma_2^{(k)} \text{LaborShare}_{jt_0} \\ & + \gamma_1^{(k)} \text{LaborShare}_{jt_0} \times \text{Post2016}_t + X'_{jt-1} \gamma^{(k)} + \tau_t^{(k)} + \delta_j^{(k)} + g^{(k)}(n_{jt}) + \nu_{jt}^{(k)} \end{aligned} \quad (11)$$

This specification tests whether import competition exerts stronger disciplinary effects in sectors where domestic market power is initially more concentrated. The same logic is extended to additional mechanism tests using interactions with baseline sectoral characteristics, such as initial markup levels, size, or revenue growth.

6 Results

6.1 First stage and identification diagnostics

I begin by assessing the relevance of the output-side instrument and the empirical distinctness of the output and input channels. Table 4 reports the first-stage estimates from regressing sectoral Chinese import penetration on the output-competition instrument Z_{jt}^{Output} , while controlling for the input-supply shifter Z_{jt}^{Input} , sector fixed effects, and year fixed effects. I report both a parsimonious specification and the baseline specification that additionally includes lagged firm-level controls. Because the treatment varies at the sector-year level, the firm-year observation count overstates the amount of independent variation; the effective design contains 609 sector-year cells.

The first-stage results show that the instrument is strongly predictive of realized import-penetration changes. In both specifications, the coefficient on Z_{jt}^{Output} is positive and precisely estimated, with an estimate of 0.892 and a standard error of 0.179. The corresponding Kleibergen–Paap rk Wald F -statistic for the excluded instrument is about 24.8 in both columns. This indicates that Chinese export growth to other high-income destinations is predictive of import competition in Turkish manufacturing sectors, although the statistic is still interpreted together with the 609 effective sector-year cells and the Rotemberg concentration reported below.

By contrast, the coefficient on the input-supply control Z_{jt}^{Input} is small and statistically insignificant in the first stage. This pattern is useful for interpretation. It suggests that, conditional on sector and year fixed effects, the main predictive power for sectoral import penetration comes from the output-side shift-share instrument rather than from the upstream input channel. In other words, the first stage is not being driven by input-cost comovement.

A related concern is whether the two shifters are too closely aligned to be interpreted as distinct channels. The table therefore also reports the small correlation between Z_{jt}^{Output} and Z_{jt}^{Input} . This positive but modest correlation indicates that the two measures are related, as one would expect given their common connection to China-driven trade shocks, but far from collinear. In the robustness, orthogonalizing the output shock with respect to the input-supply channel does not change any coefficients of interest. This supports the strategy of using Z_{jt}^{Output} as the excluded instrument while including Z_{jt}^{Input} as a control for the input-supply channel.

Table 4: First Stage and Identification Diagnostics

	Parsimonious	Baseline
	b/se	b/se
Output-competition IV	0.892*** (0.179)	0.892*** (0.179)
Input-supply control	0.164 (0.492)	0.163 (0.491)
Lagged firm controls	No	Yes
Sector FE	Yes	Yes
Year FE	Yes	Yes
Kleibergen–Paap rk Wald F	24.82	24.81
Corr. (Z_{jt}^{Output} , Z_{jt}^{Input})	0.136	0.136
Number of sectors	95	95
Effective sector-year cells	609	609
Observations	30551	30551

Standard errors in parentheses.

The reported weak-instrument statistic is the cluster-robust Kleibergen–Paap rk Wald F .

Observations are repeated firm-year rows; identification comes from the 609 sector-year treatment cells.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Finally, the near-identical first-stage estimates across the parsimonious and baseline specifications indicate that the predictive content of the instrument is not sensitive to the inclusion of lagged firm characteristics. This stability is reassuring for the subsequent IV analysis. At the same time, the combination of a moderate number of sector-year cells and concentrated Rotemberg weights implies that the estimates should be read as local to influential exposed sectors. Overall, the first-stage evidence supports the relevance of the output-side instrument and clarifies the scope of the identifying variation used in the baseline specifications.

That qualification matters quantitatively. In the Rotemberg exclusion exercise reported later, dropping the top five influential sectors reduces the Kleibergen–Paap rk Wald F -statistic from 24.8 to 3.2, and larger exclusions make the second-stage coefficient unstable in sign and magnitude. The baseline estimate is therefore best read as a disciplined local result rather than as a uniformly stable average effect for all Turkish manufacturing sectors.

6.2 Baseline IV estimates and selection diagnostic

Table 5 reports the baseline IV evidence on observed exit and markup adjustments. The estimates point to pro-competitive incumbent markup discipline in the exposed cells that carry identifying weight. In columns (2)–(4), the coefficient on instrumented import penetration is negative throughout and remains statistically significant across specifications, indicating that greater exposure to Chinese import competition reduces firms' markup adjustments. This pattern is consistent with models in which stronger competitive pressure compresses residual demand and limits firms' pricing power.

Table 5: Firm-Level Responses to Import Competition: Observed Exit and Markup Adjustments

Variable	Dependent Variable			
	Observed Exit Probability	Markup Adjustments	Markup Adjustments	Markup Adjustments
Import penetration($\widehat{\Delta IP}_{jt}$)	0.914** (0.459)	-0.422* (0.229)	-0.462** (0.226)	-0.454* (0.253)
Input supply shocks(Z_{jt}^{Input})	0.878 (1.078)	0.432 (0.490)	0.517 (0.470)	0.477 (0.486)
Specification				
Lagged Controls	No	No	Yes	Yes
Selection Diagnostic	No	No	No	Yes
Fixed Effects		Sector and Year FE		
KP rk Wald F	33.969	24.813	24.777	25.105
Observations	30,591	31,403	30,551	30,445
R^2	0.015	-0.003	-0.002	0.078

Standard errors clustered at ISIC4 sector level in parentheses.

Column (1) reports marginal effects from the observed-exit specification. Columns (2)–(4) estimate markup adjustments with increasing controls. KP rk Wald F denotes the Kleibergen–Paap weak-instrument statistic for the excluded instrument.

Chinese import penetration is instrumented with shift-share measure Z_{jt}^{Output} based on Chinese exports to high-income countries.

Observed exit denotes disappearance from ORBIS/AMADEUS between $t - 1$ and t and should not be interpreted as validated legal firm death.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

At the same time, column (1) shows that import competition also affects observed survival in the panel. The positive and significant coefficient in the observed-exit regression indicates that sectors more exposed to Chinese import penetration experience more

sample disappearance. This matters for identification because the observed sample of markup changes is conditional on survival. If import competition disproportionately removes firms with systematically different markup dynamics, conventional IV estimates on surviving firms may conflate within-firm pricing responses with composition effects arising from endogenous selection.

To address this issue, column (4) implements the selection diagnostic based on a control function in the spirit of Backus (2020), using lagged EBIT margin as a proxy for the latent firm state governing observed survival and subsequent adjustment decisions. The diagnostic estimate remains negative and significant, implying that the decline in markup adjustments is not solely an artifact of selection on this observed state proxy. Hence, even after allowing for this reduced-form survival adjustment, the evidence supports a within-firm disciplining effect of import competition on markups.

Comparing columns (3) and (4) further suggests that selection on the EBIT-margin proxy is not dominant. Once lagged firm controls are included, the coefficient on import penetration is -0.462 ; after adding the polynomial control-function diagnostic, it becomes -0.454 . The stability of the estimate indicates that this observed survival proxy does not overturn the baseline conclusion, although it remains important to track selection explicitly when panel disappearance responds to the same trade shock. Put differently, Chinese import competition appears to reduce markup growth among surviving firms, while also reshaping which firms remain observed in the sample.

Finally, the coefficient on the input-supply shock is statistically insignificant in all markup specifications. This suggests that, in this setting, the output-competition channel is the main force behind firm-level pricing adjustment, whereas upstream input-supply effects do not generate a robust independent effect on markup changes. I return to the input-supply shifter in Section 8, where it is used as a mechanism and decomposition device rather than as a second headline causal result.

6.3 From firm to sector

Table 6 connects the firm-level evidence to sector-level markup dynamics by decomposing changes in inverse sector markups into four components: within-firm adjustment, reallocation across continuing firms, observed entry, and observed exit. The sectoral markup is constructed as the inverse of the market-share-weighted average of firm-level inverse markups, so that changes in sectoral market power reflect both individual pric-

ing responses and changes in the composition of firms and market shares. This aggregation follows the harmonic-mean logic emphasized in the recent markup literature.

Table 6: Decomposition of Sector-Level Markup Adjustments: Comparison Specification

Variable	Dependent Variable				
	Total	Within	Reallocation	Entry	- Exit
Panel A: Regression coefficients					
Import Penetration Change ($\widehat{\Delta IP_{jt}}$)	0.225 (0.441)	0.088 (0.170)	-0.191 (0.728)	-0.371 (0.430)	0.663 (0.571)
Input Supply Shocks (Z_{jt}^{Input})	3.356** (1.661)	-0.764 (0.465)	8.055** (3.399)	-1.375 (2.556)	-3.331 (2.543)
Observations	609	610	610	610	610
Panel B: Shares of gross absolute decomposition mass (size-weighted, %)					
Within-firm adjustment			48.44		
Reallocation			49.30		
Entry			1.16		
Exit			1.09		
Total			100.00		

Standard errors in parentheses.

Panel A reports decomposition regressions under the comparison specification.

Panel B reports shares of *gross absolute* decomposition mass attributable to each component.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A shows that, under the sector-level IV specification, the estimated effect of output import competition on the total change in sector-level markups is small and statistically insignificant. The same is true for each individual component. In particular, the coefficient on within-firm adjustment is close to zero, while the reallocation and observed-entry margins are negative and the observed-exit margin is positive, but all are imprecisely estimated. Taken together, these estimates suggest that, once firm-level responses are aggregated to the sector level, the pro-competitive effect of Chinese import penetration does not translate into a precisely estimated change in sectoral markups. By contrast, the input-supply shock is associated with a significant increase in inverse

sector markups, which is equivalent to lower sectoral markup growth. This effect is driven almost entirely by the reallocation component: sectors more exposed to upstream input shocks experience a statistically significant shift in market shares toward firms with higher inverse markups, or lower markups. The within-firm, observed-entry, and observed-exit terms are all statistically insignificant.

Panel B shows that, in gross absolute terms, sectoral markup adjustment is overwhelmingly accounted for by the within-firm and reallocation margins. The within component represents 48.44% of total absolute adjustment, while reallocation contributes 49.30%. Observed entry and observed exit are quantitatively negligible, each accounting for about 1% of the total. Thus, the accounting decomposition makes clear that sector-level markup movements are shaped almost entirely by incumbent firms, through a combination of their own markup changes and the redistribution of market shares across survivors.

Overall, the table highlights an important difference between firm-level and sector-level responses. At the firm level, import competition disciplines markups among surviving firms. At the sector level, however, this effect is partly offset by countervailing reallocation forces, so the total effect becomes weak. A consistent pattern across both levels is that observed entry and exit do not play a major role, and most markup variation comes from continuing firms.

6.4 Distributional heterogeneity

Average effects can conceal substantial heterogeneity in firms' responses to import competition. To examine how the trade shock affects different parts of the markup-adjustment distribution, I estimate grouped IV quantile regressions at the sector-year level. The results reveal a clear asymmetric pattern. Across the central part of the distribution, from the 10th to roughly the 70th percentile, the estimated effects of Chinese import competition on markup adjustments are close to zero. By contrast, the coefficients become increasingly negative from the upper quartile onward, reaching about -2.0 at the 90th percentile. Figure 7 shows that the pro-competitive effect is therefore concentrated in the upper tail of the distribution rather than spread uniformly across firms.

This pattern suggests that import competition primarily constrains large positive markup adjustments among firms located at the top of the markup-growth distribution. In other words, the main within-firm discipline effect does not operate through

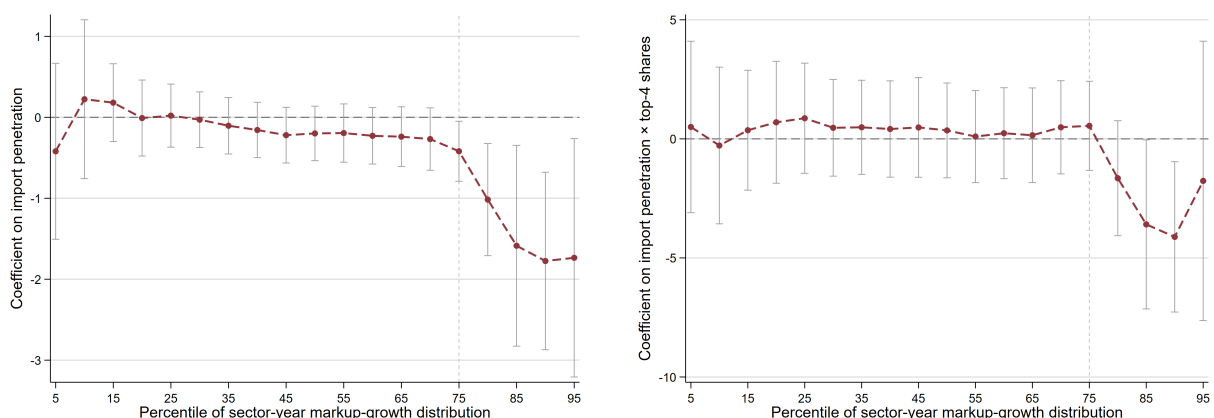


Figure 7: Point estimates and 95% confidence intervals across deciles of the markup-adjustment distribution. The left panel reports the baseline grouped IV quantile estimates for $\beta_O^{(k)}$. The right panel reports the interaction coefficients by pre-period domestic concentration (CR4), showing that the negative upper-tail effects are stronger in more concentrated sectors.

a broad compression of all firms' pricing behavior, but through a disproportionate reduction in upper-tail markup growth. Firms in the center of the distribution, which are already making small or moderate markup adjustments, appear much less responsive to the trade shock. This finding is consistent with residual-demand discipline, but it should not be read mechanically as a statement that the baseline quantile exercise ranks firms by markup levels. The appendix check that ranks firms by lagged markup levels provides the closer test of that level-based interpretation.

I next examine whether this upper-tail compression depends on domestic market structure. To do so, I interact import penetration with a measure of pre-period sectoral concentration, proxied by the domestic CR4 ratio. The estimates, reported in Figure 7, show that the negative upper-tail effects are stronger in more concentrated sectors. The interaction coefficients remain close to zero in the lower and central quantiles, but become increasingly negative at high quantiles when concentration is high. This indicates that the compression of the markup-adjustment distribution is amplified in markets where a small number of firms account for a large share of domestic sales.

Taken together, these results support a mechanism in which import competition compresses the largest positive markup adjustments, especially in concentrated domestic industries. The evidence is therefore more consistent with targeted compression at the top

of the markup-growth distribution than with a uniform decline in markups across all firms. This also helps reconcile the firm-level and sector-level results: even when average sectoral effects appear modest, import competition may still exert economically meaningful pro-competitive pressure by limiting upper-tail markup growth in the exposed cells that carry identifying weight.

7 Threats to Identification and Robustness

This section asks whether the two headline empirical claims survive the most relevant threats to interpretation. The first claim is that Chinese final-goods competition reduces markup growth among observed incumbent firms. The second is that this discipline is concentrated in the upper tail of the markup-adjustment distribution. I organize the checks around four questions. First, is the IV result driven by invalid pre-trends, omitted demand shocks, or a few influential sectors? Second, does the result depend on how markups or the firm sample are constructed? Third, is the upper-tail result an artifact of quantile construction? Fourth, can a narrow oligopoly mechanism reproduce the same IV moments and absorb the residual output-shock variation? The ordering is intentional: the empirical identification comes first, and the model-based validation is auxiliary.

7.1 Identification: exposure, pre-trends, and influential sectors

The first set of checks addresses the identifying variation behind the baseline IV. Table 11 shows that the main coefficient is not driven by a narrow control set or functional-form choice. The preferred estimate is -0.462 and remains statistically significant under wild-cluster bootstrap inference. Removing lagged firm controls one block at a time leaves the coefficient close to the baseline: the estimates range from -0.462 with the preferred controls to -0.396 with no controls. Excluding the input-supply control also leaves a negative coefficient. The quadratic specification is less informative because the linear and squared terms are imprecise, but it provides no evidence that the baseline is hiding a strong nonlinear pattern.

The main identification caveat is not instability across controls; it is the concentration of identifying weight. Figure 5 and Appendix Table 12 show that the largest Rotemberg weights come from small but highly China-exposed cells. Table 15 confirms that these

cells are central to the first stage: dropping the five largest-weight sectors lowers the Kleibergen–Paap F -statistic from 24.8 to 3.2, and broader exclusions make the second stage unstable. This does not invalidate the estimate, but it defines the estimand. The baseline result should be read as a local effect for influential exposed sectors, not as a homogeneous effect for all Turkish manufacturing. As an additional visual diagnostic, the synthetic-control exercise (Abadie, 2021) for ISIC4 2790 shows a post-treatment deviation consistent with lower markup growth relative to a weighted counterfactual of lower-exposure sectors.

Two checks support the timing and exclusion assumptions on this support. Appendix Table 14 shows that future output-side China shocks do not predict pre-period trends in markups, revenue, or concentration. Appendix Table 13 adds high-income global-demand and Turkish export-demand controls; the import-penetration coefficient remains close to the baseline. Finally, because output-market and input-supply exposure are both built from China’s trade growth, Appendix Tables 16–19 re-estimate the main specifications after residualizing Z_{jt}^{Output} with respect to Z_{jt}^{Input} . The markup coefficient remains -0.462 , with the same standard error and first-stage strength. The result therefore comes from the output-competition component of China exposure rather than from linear overlap with the upstream input channel.

7.2 Markup measurement and sample selection

The second set of checks asks whether the baseline result depends on the markup measure or on the composition of the observed firm panel. Table 20 implements the Kirov et al. (2026) markup procedure as an alternative to the baseline production-based measure. This check is demanding because it requires lagged markup shifters and a Markov process for revenue productivity, which substantially reduces the usable sample. I therefore treat the estimates as illustrative rather than decisive. Even so, the coefficients remain negative and similar in magnitude.

Table 21 shows that alternative trimming and winsorization rules preserve the sign, broad magnitude, and upper-tail concentration of the main result. Table 22 then restricts the sample to firms observed for longer spells. The coefficient attenuates and becomes less precise as the panel becomes more selective. This is the expected direction: conditioning on long observed histories removes part of the firm-composition margin through which import competition changes the observed set of incumbents. I therefore keep the

full unbalanced panel as the preferred sample and interpret long-lived-firm estimates as conservative composition checks.

A final measurement concern is that value-based import penetration may partly capture concentration in Chinese supply within narrow products rather than competitive pressure from China as a whole. Appendix Tables 26–29 add product-level concentration in Chinese imports within ISIC4-year cells. The baseline IV coefficient is unchanged. Interactions with Chinese product concentration are imprecise, and the triple-interaction specifications are weakly identified. The data therefore provide no evidence that the main result is mechanically driven by granular Chinese product supply.

7.3 Distributional design

The third set of checks protects the upper-tail claim. The baseline grouped-IV quantile exercise ranks firms by markup adjustment within sector-year cells, so it could be sensitive to binning, ranking choices, or small-cell order statistics. Appendix Figures 10–12 address these concerns directly. Quintiles reproduce the same broad shape as deciles. Ranking firms by lagged markup levels, rather than by current markup growth, still produces stronger negative effects near the top. Excluding thin sector-year cells leaves the upper-tail pattern visible.

The Monte Carlo evidence explains why the main specifications control for cell size and why I treat the upper tail with extra caution. In a null design where sector-year cell size is correlated with the instrument, the uncorrected grouped-IV quantile estimator over-rejects heavily; Table 24 reports a false rejection rate of 0.694 at Q90. Adding a cubic polynomial in sector-year cell size lowers the Q90 rejection rate to 0.152. The targeted spline-bootstrap correction in Table 25 lowers it further to 0.050. These diagnostics support the baseline choice to control flexibly for cell size and to report the spline-bootstrap version as an upper-tail robustness check. The substantive conclusion is that the upper-tail compression is not a simple artifact of decile construction or thin cells.

7.4 Indirect-inference validation of the output-competition channel

The final check asks whether a parsimonious oligopoly mechanism can internally reproduce the IV evidence. I estimate a causal-core model by simulated method of moments, equivalently a Wald indirect-inference estimator. Let $\hat{m}$ denote the vector of auxiliary

causal moments estimated in the data, and let $m^s(\theta)$ denote the same moments computed from the simulated panel under structural parameter vector θ . The estimator solves

$$\hat{\theta} = \arg \min_{\theta \in \Theta} [\hat{m} - m^s(\theta)]' W [\hat{m} - m^s(\theta)]. \quad (12)$$

The exercise validates the output-competition mechanism among incumbent firms, not a full sector-aggregation or welfare model. The five targeted moments are the baseline firm-level IV response, the q80, q85, and q90 grouped-IV responses, and the q90 interaction with sector concentration. They discipline the sign, magnitude, upper-tail shape, and concentration gradient of the incumbent markup response. Sector inverse markups, input-shock responses, exit and selection outcomes, allocative-wedge objects, and within/between/entry/exit decompositions remain diagnostic and do not enter the criterion.

Table 7: Causal-core parameter estimates

Parameter	Estimate	Interpretation
Output scale	2.138	Scale of the output-competition channel
Output-markup effect	-0.329	Baseline output-shock effect on markup growth
High-markup interaction	-2.102	Extra effect for high-markup units
Concentration interaction	-2.222	Extra effect in concentrated markets
Share-dynamics effect	-0.235	Output-shock component in share dynamics
Mean reversion	1.025	Auxiliary dynamic adjustment parameter
Drift	0.350	Average residual drift in markup dynamics

Notes: The input and selection blocks are fixed at zero. The estimates therefore describe the output-competition mechanism, not input pass-through or endogenous exit.

Table 7 has the expected signs. Output competition lowers markup growth on average, and the effect becomes stronger among high-markup units and in concentrated markets. The estimated share-dynamics term is also negative, consistent with exposed incumbents losing ground when foreign output-market pressure rises.

The mean-reversion parameter should be read as an auxiliary dynamic adjustment parameter inside the indirect-inference approximation, not as a structural persistence estimate from the markup literature. Its estimated value of 1.025 helps match the level and

shape of the incumbent IV response across the baseline, upper-tail, and concentration-gradient moments.

Table 8: Targeted causal moment fit

Target moment	Data	Model	Assessment
Baseline IV, $\Delta \ln \mu$	-0.429	-0.414	Very close central response
Grouped IV, q80	-0.803	-0.924	Correct sign; slightly too strong
Grouped IV, q85	-1.236	-1.009	Correct sign; modestly smaller
Grouped IV, q90	-1.625	-1.572	Very close upper-tail response
Grouped IV, q90 $\times$ CR4	-5.371	-5.232	Very close concentration gradient

Table 8 shows that the model matches the central causal evidence. The baseline IV coefficient, the q90 grouped-IV effect, and the concentration gradient are all closely reproduced. The q80–q85 fit is not exact, but the model clearly generates the same upper-tail discipline found in the data. The minimized criterion is $Q = 1.099$.

The residual-IV validation is the strongest external check. I subtract the model-implied output-competition component from incumbent markup growth and re-estimate the IV regression on the residual. The excluded output shifter no longer predicts a meaningful residual markup response: the coefficient is -0.052 , with standard error 0.295 , $31,375$ observations, 90 sector clusters, and a first-stage F -statistic of 22.93 . The model therefore absorbs the output-shock variation it was built to explain, and the validation is not driven by a weak first stage.

The sector inverse-markup regression is best read as a secondary aggregation diagnostic. Its empirical IV estimate is less precise and is not consistently significant across the causal specifications, so I do not use it as a core target moment. This keeps the structural validation tied to the empirically strongest result: output competition lowers markup growth among observed incumbent firms, especially in the upper tail and in more concentrated sectors. Appendix B.3.11 reports the quantitative counterfactuals, input and exit diagnostics, optimization details, and non-targeted moments.

Taken together, the checks strengthen the central result while narrowing its interpretation. Chinese import competition lowers markup growth among observed incumbents in the exposed cells that carry the IV, and the effect is concentrated in the upper tail of the markup-adjustment distribution. The evidence does not support a claim of a uniform markup decline across all manufacturing sectors; it supports a more precise claim of local pro-competitive discipline in the sectors most exposed to the China shock.

8 Mechanism evidence and interpretation

The mechanism evidence asks whether the baseline output-competition result reflects residual-demand discipline, input-cost relief, reallocation among continuing firms, or non-price adjustment. I keep the tables in Appendix B.4. The goal is interpretive: each exercise tests whether the mechanisms line up with the incumbent and sector-level results above.

8.1 State dependence and residual-demand evidence

To test residual-demand discipline, I interact instrumented import penetration with lagged firm markups and compare the implied marginal effects with the grouped-IV quantile evidence. Appendix Table 34 shows a significant interaction and negative marginal effects at the 25th, 50th, and 75th percentiles. The stronger result is therefore not a monotone high-markup story; it is the upper-tail compression documented in Section 6.4.

8.2 Input-cost relief and reallocation among continuing firms

To separate cost relief from output discipline, I interact the input-supply shifter and output import penetration with pre-period Chinese-input dependence. Appendix Table 35 shows a positive input-shock interaction in the markup regression, while the output interaction is imprecise. This supports a cost-relief channel, not a second residual-demand result.

I then test whether the shocks reallocate market share toward large or dominant incumbents. Appendix Table 36 shows no significant output-shock reallocation toward large firms, while input-shock interactions are negative for size, large-firm status, and CR4 membership. This matches the sectoral decomposition: output competition mainly

disciplines continuing firms, while input exposure reallocates shares among continuers without strengthening dominant survivors.

8.3 Auxiliary outcomes

As a final check, I estimate the same mechanism specifications for revenue, domestic sales, and exports. Appendix Table 37 shows that import competition lowers revenue and domestic-sales growth, while export effects are imprecise. This supports the earlier interpretation: markup discipline comes with domestic demand pressure, not systematic export redirection.

9 Conclusion

This thesis studies how Chinese import competition reshapes market power in Turkish manufacturing. The central design is an output-competition shift-share IV based on Chinese exports to other high-income markets. Using ORBIS/AMADEUS firm data, I find evidence of pro-competitive incumbent markup discipline in high-weight exposed cells: higher Chinese import penetration reduces markup growth among surviving firms, with stronger effects in the upper tail of the markup-adjustment distribution and in more concentrated sectors. A causal-core indirect-inference exercise reproduces the main IV moments and passes a residual-IV validation, which supports the output-competition mechanism without turning the exercise into a full welfare model. The input-supply extension shows that Chinese input exposure does not robustly lower within-firm markups, but it helps explain market-share reallocation in sectors more exposed to Chinese inputs.

The welfare interpretation is cautiously pro-competitive. Lower markup growth among incumbents means that Chinese competition reduces price-cost wedges for surviving Turkish manufacturers, especially where market power is initially most relevant. If lower markups are passed through to buyers, this points toward gains for Turkish consumers and downstream firms through lower prices or slower price growth. The sectoral decomposition is treated as an accounting diagnostic rather than as a targeted structural object: it helps describe where inverse-markup changes appear in the observed data, but its component shares can move with accounting definitions, sample windows, and the treatment of observed entry and exit. Small sales-weighted inverse-markup contributions of entry and exit therefore do not imply that firm-count turnover is economically

irrelevant. The thesis does not observe transaction prices, wages, or household welfare directly. For wage earners, the implications are ambiguous: stronger competition can discipline rents and benefit consumers, but the associated declines in domestic sales and observed panel survival may also weaken labor demand in exposed sectors.

The aggregate interpretation is narrower than a broad claim about “creative destruction.” Import competition raises observed exit probabilities in the panel, yet the selection diagnostic shows that the within-incumbent disciplining effect is not driven by the EBIT-margin survival proxy. At the sector level, total markup effects are weak because within-firm discipline is partly offset by reallocation across continuing firms. The indirect-inference counterfactuals reinforce this local interpretation: removing the fitted output-competition channel has negligible aggregate effects, even though sector-year effects can be large in concentrated markets. Since ORBIS disappearance is not validated firm death, the safest conclusion is that Chinese import competition disciplines markup growth among observed incumbents and reshuffles market shares among survivors; it is not, by itself, definitive evidence on true firm death or entry.

These findings have policy implications for the relationship between trade policy and domestic competition policy, but they do not imply that trade policy is a full substitute for antitrust enforcement. Trade shocks discipline firms unevenly, the identifying variation is concentrated in exposed sector-year cells, and many non-tradable or institutionally protected markets remain outside the reach of import competition. Input liberalization can also work differently from final-goods liberalization: easier access to imported intermediates may improve efficiency and reallocate activity without necessarily compressing each firm’s markup.

This thesis has three main limitations. First, the estimates are local to the sector-year cells carrying the identifying weight, so they should not be interpreted as a uniform effect across all Turkish manufacturing. Second, markup measurement remains sensitive to accounting data quality, production-function assumptions, and the choice of flexible input. Third, the current structural layer switches off input and selection mechanisms, so it validates the output-competition channel but cannot support welfare or long-run composition claims. The most immediate extension would link ORBIS/AMADEUS to administrative firm registries, social-security records, or customs transactions to separate legal exit, reporting attrition, employment adjustment, and input importing. A second extension would match the firm panel to product-level price data in order to estimate

pass-through and quantify consumer welfare rather than inferring it from markups.

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A Theoretical Appendix

This appendix records the derivations that are useful for interpreting the empirical results but too mechanical for the main text. It focuses on three objects used in Section 2: the perceived elasticity under nested CES demand, the corresponding Cournot markup and sectoral aggregation formulas, and the dynamic exit cutoff.

A.1 Perceived elasticity under nested CES demand

Let $\tilde{s}_{ij}$ denote firm i 's expenditure share within the domestic segment of sector j , and let s_{ij} denote its share in total sectoral expenditure. Under the nested CES structure in Section 2, two derivative identities are central:

$$\frac{\partial \ln Q_j^D}{\partial \ln q_{ij}} = \tilde{s}_{ij}, \quad \frac{\partial \ln Q_j}{\partial \ln q_{ij}} = s_{ij}. \quad (13)$$

The first identity says that a firm's output matters more for the domestic composite when that firm is large inside the domestic segment. The second says that the effect on the entire sectoral composite is the firm's total sectoral expenditure share.

Using the inverse demand for a domestic variety,

$$\ln p_{ij} = \ln P_j^D - \frac{1}{\rho_j} \ln q_{ij} + \frac{1}{\rho_j} \ln Q_j^D + \frac{1}{\rho_j} \ln \xi_i,$$

and differentiating with respect to $\ln q_{ij}$ yields

$$\frac{d \ln p_{ij}}{d \ln q_{ij}} = \frac{d \ln P_j^D}{d \ln q_{ij}} - \frac{1 - \tilde{s}_{ij}}{\rho_j}. \quad (14)$$

The middle nest implies

$$\frac{d \ln P_j^D}{d \ln q_{ij}} = \frac{d \ln P_j}{d \ln q_{ij}} - \frac{\tilde{s}_{ij} - s_{ij}}{\nu_j}, \quad (15)$$

while the top nest implies

$$\frac{d \ln P_j}{d \ln q_{ij}} = -\frac{s_{ij}}{\eta}. \quad (16)$$

Combining (14)–(16) gives the inverse-demand elasticity

$$-\frac{d \ln p_{ij}}{d \ln q_{ij}} = \frac{1 - \tilde{s}_{ij}}{\rho_j} + \frac{\tilde{s}_{ij} - s_{ij}}{\nu_j} + \frac{s_{ij}}{\eta}. \quad (17)$$

Define the perceived elasticity under Cournot competition by

$$\frac{1}{\sigma_{ij}^C} \equiv -\frac{d \ln p_{ij}}{d \ln q_{ij}}.$$

Then

$$\sigma_{ij}^C = \left(\frac{1 - \tilde{s}_{ij}}{\rho_j} + \frac{\tilde{s}_{ij} - s_{ij}}{\nu_j} + \frac{s_{ij}}{\eta} \right)^{-1}. \quad (18)$$

Equation (18) is the key theoretical object behind Prediction 2: when a firm is more important inside broader and broader aggregates, the residual demand it faces becomes less elastic.

A.2 Cournot markup and sectoral aggregation

Firm i solves

$$\max_{q_{ij}} \pi_{ij} = (p_{ij}(q_{ij}, q_{-ij}) - \text{mc}_{ij})q_{ij} - f_{ij}.$$

The first-order condition is

$$p_{ij} + q_{ij} \frac{\partial p_{ij}}{\partial q_{ij}} - \text{mc}_{ij} = 0. \quad (19)$$

Divide (19) by p_{ij} and use the definition of σ_{ij}^C :

$$1 - \frac{1}{\sigma_{ij}^C} = \frac{\text{mc}_{ij}}{p_{ij}}.$$

Defining the markup $\mu_{ij} \equiv p_{ij}/\text{mc}_{ij}$ gives

$$\frac{1}{\mu_{ij}} = 1 - \frac{1}{\sigma_{ij}^C}, \quad \mu_{ij} = \frac{\sigma_{ij}^C}{\sigma_{ij}^C - 1}. \quad (20)$$

Substituting (18) yields the inverse-markup condition used in the main text:

$$\frac{1}{\mu_{ij}} = \left(1 - \frac{1}{\rho_j} \right) - \left[\frac{1}{\lambda_j^D} \left(\frac{1}{\nu_j} - \frac{1}{\rho_j} \right) + \left(\frac{1}{\eta} - \frac{1}{\nu_j} \right) \right] s_{ij}, \quad (21)$$

where λ_j^D is the realized domestic expenditure share. The main text writes this object with Λ_j^D for expositional simplicity. The important point is that the expression is linear in *inverse markups*, not in markups themselves.

Sectoral markup is the market-share-weighted harmonic mean,

$$\mu_j \equiv \left(\sum_{i=1}^{N_j} \frac{s_{ij}}{\mu_{ij}} \right)^{-1}.$$

Taking inverses and substituting (21) gives

$$\frac{1}{\mu_j} = \left(1 - \frac{1}{\rho_j} \right) - \left[\frac{1}{\lambda_j^D} \left(\frac{1}{\nu_j} - \frac{1}{\rho_j} \right) + \left(\frac{1}{\eta} - \frac{1}{\nu_j} \right) \right] \text{HHI}_j, \quad (22)$$

where $\text{HHI}_j = \sum_i s_{ij}^2$. Under $\rho_j > \nu_j > \eta$, the bracketed term is positive, so sectoral markups rise with concentration. This is the theoretical basis for the concentration interaction in the grouped-IV quantile evidence.

A.3 Dynamic exit cutoff

Let x_{ijt} denote the firm's state, including productivity, lagged market share, and the output- and input-trade shocks. If the firm stays active, its value is

$$V(x_{ijt}) = \max \{0, \pi(x_{ijt}) + \beta \mathbb{E}[V(x_{ij,t+1}) \mid x_{ijt}]\}. \quad (23)$$

Define the continuation surplus

$$W(x_{ijt}) \equiv \pi(x_{ijt}) + \beta \mathbb{E}[V(x_{ij,t+1}) \mid x_{ijt}].$$

Then the optimal exit rule is

$$\text{Exit}_{ijt} = 1 \iff W(x_{ijt}) < 0. \quad (24)$$

Equivalently, if profitability can be summarized by a latent scalar index θ_{ijt} , there exists a cutoff $\theta^*(x_{ijt})$ such that

$$\text{Exit}_{ijt} = 1 \iff \theta_{ijt} < \theta^*(x_{ijt}). \quad (25)$$

This is all the main text needs from the dynamic extension. Trade shocks affect both current profits and continuation values, so the sample of surviving firms is not random. The empirical selection correction in Section 4.2 is therefore a reduced-form way to purge the markup equation of selection on unobservables, not a structural estimation of (25).

A.4 From the static model to the first-difference specification

This appendix makes explicit the bridge between the static oligopoly model and the first-difference IV specification used in the empirical sections. The derivations above show why inverse markups are the natural theoretical object. The remaining step is to interpret annual changes in estimated log markups as a local approximation to movements between adjacent equilibria.

Local first-difference approximation

Let the equilibrium markup of firm i in sector j and year t be written as

$$\mu_{ijt} = M_j(\Lambda_{jt}^D, s_{ijt}, mc_{ijt}; \theta_j),$$

where Λ_{jt}^D is the domestic expenditure share, s_{ijt} is the relevant domestic market-share state, mc_{ijt} is marginal cost, and θ_j collects demand and technology parameters. The closed-form markup condition in Section 3 is written in inverse-markup form. Since $\log \mu_{ijt} = -\log \mu_{ijt}^{-1}$,

$$\Delta \log \mu_{ijt} = -\Delta \log \mu_{ijt}^{-1} \approx -\frac{\Delta \mu_{ijt}^{-1}}{\mu_{ij,t-1}^{-1}} = -\mu_{ij,t-1} \Delta \mu_{ijt}^{-1}.$$

Thus, for small annual changes, the sign predictions for changes in inverse markups translate directly into sign predictions for changes in log markups, with the opposite sign.

Define the local state vector

$$z_{ijt} = (\log \Lambda_{jt}^D, s_{ijt}, \log mc_{ijt}).$$

A first-order expansion of $\log M_j$ around the lagged equilibrium $z_{ij,t-1}$ gives

$$\Delta \log \mu_{ijt} \approx \psi_{\Lambda,ij,t-1} \Delta \log \Lambda_{jt}^D + \psi_{s,ij,t-1} \Delta s_{ijt} + \psi_{c,ij,t-1} \Delta \log mc_{ijt} + r_{ijt},$$

where the ψ -terms are local derivatives and r_{ijt} collects higher-order terms. The empirical coefficient on a trade variable should therefore be read as a local reduced-form equilibrium response, not as a primitive structural parameter. It combines the direct residual-demand effect of the shock with any induced movement in market shares and marginal costs that remains after controls are included.

Mapping to the estimating equation

The empirical design uses two observable trade-exposure measures to proxy for the theoretical state variables. Output-side Chinese import penetration proxies for changes in the domestic expenditure share:

$$\Delta \log \Lambda_{jt}^D = \kappa_O \Delta IP_{jt} + \nu_{jt}^O, \quad \kappa_O < 0.$$

The input-supply shifter proxies for upstream cost exposure:

$$\Delta \log mc_{ijt} = \kappa_I Z_{jt}^{\text{Input}} + \nu_{ijt}^I.$$

Substituting these proxy mappings into the local expansion yields the baseline empirical analogue

$$\Delta \log \hat{\mu}_{ijt} = \beta_O \Delta IP_{jt} + \beta_I Z_{jt}^{\text{Input}} + X'_{ijt-1} \gamma + \delta_j + \tau_t + \varepsilon_{ijt}.$$

Here β_O is the local response of markup growth to output-market import competition, conditional on the input channel, lagged firm controls, sector fixed effects, and year fixed effects. The input coefficient β_I captures the separate reduced-form association between upstream exposure and markup growth.

Because ΔIP_{jt} may respond to Turkish sectoral shocks, it is instrumented with the output-side shift-share variable

$$Z_{jt}^{\text{Output}} = s_{j,\text{CHN}}^{2009} g_{jt},$$

where $s_{j,\text{CHN}}^{2009}$ is baseline Turkish exposure to Chinese imports and g_{jt} is Chinese export growth to other high-income markets in sector j . The input control is constructed from the same output shocks using predetermined input-output links,

$$Z_{jt}^{\text{Input}} = \sum_k w_{jk}^{2003} Z_{kt}^{\text{Output}}.$$

This separates the excluded product-market competition shock from the observed upstream exposure channel. The resulting IV estimate is local to the sector-year cells that carry identifying weight in the shift-share design.

Selection and aggregation

The first-difference outcome is available only when a firm reports the required accounting variables in adjacent years. The dynamic selection block of the model implies that survival can depend on latent profitability states that also affect markup adjustment. The corrected specification

therefore adds a control function in lagged EBIT margin, $a_{i,t-1}$:

$$\Delta \log \hat{\mu}_{ijt} = \beta_O \widehat{\Delta IP}_{jt} + \beta_I Z_{jt}^{\text{Input}} + h(a_{i,t-1}) + X'_{ijt-1} \gamma + \delta_j + \tau_t + \epsilon_{ijt}.$$

In the main text, $h(\cdot)$ is implemented as a third-order polynomial. Under the maintained control-function assumption, β_O is a within-incumbent response among observed survivors. It should not be interpreted as the total effect of import competition on all firms, including those that disappear from the panel.

The same logic explains why the sectoral analysis is written in inverse markups. Under the harmonic-mean sectoral markup, inverse markups aggregate linearly, so sectoral changes can be decomposed into within-firm adjustment, reallocation across continuing firms, observed entry, and observed exit. This decomposition is an accounting bridge between the incumbent-level IV estimates and sector-level markup movements; it does not impose that the average firm response and the aggregate response have the same sign.

Assumptions and scope

The bridge rests on four maintained conditions. First, annual changes must be small enough for the Taylor remainder to be absorbed into the residual. Second, sectoral demand and technology parameters must be stable enough that the local derivatives do not drift sharply within the estimation window. Third, the observable import-penetration and input-supply measures must be informative proxies for the theoretical domestic-share and cost states. Fourth, conditional on fixed effects and controls, Chinese export growth to other high-income markets must be orthogonal to residual Turkish sector-year shocks affecting markups.

These conditions define the scope of the empirical interpretation. The design identifies relative exposure effects across Turkish manufacturing sectors, not the common aggregate effect of China's rise on markups, prices, or welfare. The coefficient is also weighted toward influential sector-year cells, as emphasized by the Rotemberg-weight diagnostics in Section 5.1. Within that scope, the model justifies interpreting a negative coefficient on instrumented import penetration as evidence of markup discipline among observed incumbents.

B Empirical Appendix

This appendix follows the order of the empirical sections in the main text. It first documents data construction and markup measurement, then reports supporting estimates for the empirical strategy and main results, then places robustness checks before the mechanism tables. Each

table is included to support a specific claim in the main text rather than to introduce a separate headline result.

B.1 Data and Measurement

B.1.1 Markup estimation

Baseline estimator: I estimate firm markups using the production approach in De Loecker and Warzynski (2012), with the implementation guided by De Ridder et al. (2026). The baseline uses COGS as the flexible input because it is the most consistently observed short-run variable input in ORBIS/AMADEUS. For firm i in year t , let

$$\begin{aligned} r_{it} &= \log R_{it}, \\ x_{it} &= \log X_{it}, \\ k_{it} &= \log K_{it}, \end{aligned}$$

where R_{it} is revenue, X_{it} is COGS, and K_{it} is capital. The estimation sample requires $R_{it} > 0$, $X_{it} > 0$, and $K_{it} > 0$.

Markup recovery: For a flexible input, the first-order condition implies

$$\mu_{it} = \theta_{it}^{\text{COGS}} \frac{R_{it}}{X_{it}},$$

where $\theta_{it}^{\text{COGS}}$ is the output elasticity of COGS. In logs, the empirical markup is therefore

$$\log \hat{\mu}_{it} = \log \hat{\theta}_{it}^{\text{COGS}} + \log R_{it} - \log X_{it}.$$

The main empirical outcome is $\Delta \log \hat{\mu}_{it}$ rather than the level $\log \hat{\mu}_{it}$. This choice is deliberate: markup levels inferred from accounting data are more exposed to persistent reporting differences, production-function misspecification, and denominator error, whereas within-firm changes are better aligned with the time variation in the import-competition shock.

Cobb–Douglas elasticity: In the baseline implementation, I estimate the following revenue production function separately within each ISIC4 sector:

$$r_{it} = \beta_x x_{it} + \beta_k k_{it} + \lambda_t + \varepsilon_{it}.$$

The Cobb–Douglas specification imposes a constant COGS elasticity within sector, so

$$\hat{\theta}_{it}^{\text{COGS}} = \hat{\beta}_x.$$

Translog sensitivity: As a sensitivity check, I allow the flexible-input elasticity to vary with input levels by estimating a two-input translog specification:

$$r_{it} = \beta_x x_{it} + \beta_k k_{it} + \beta_{xx} x_{it}^2 + \beta_{kk} k_{it}^2 + \beta_{xk} x_{it} k_{it} + \lambda_t + \varepsilon_{it}.$$

The implied firm-year COGS elasticity is then

$$\hat{\theta}_{it}^{\text{COGS}} = \frac{\partial r_{it}}{\partial x_{it}} = \hat{\beta}_x + 2\hat{\beta}_{xx} x_{it} + \hat{\beta}_{xk} k_{it}.$$

B.1.2 Data details

ORBIS/AMADEUS firm panel.

The core microdata come from Bureau van Dijk company accounts. The project archive uses two raw Turkish extracts: a financial accounting file and a BvD industry classification file. The financial file contains harmonized balance-sheet, income-statement, and ratio variables, including operating revenue, costs of goods sold, tangible and intangible fixed assets, total assets, export revenue, debt, current assets and liabilities, working capital, interest expenses, cash holdings, EBIT margins, gross margins, and liquidity ratios. The classification file provides BvD firm identifiers and NACE Rev. 2 sector codes. I merge the two files by the BvD identifier and then retain the Turkish manufacturing panel after sector and deflator matches.

The cleaning procedure is designed to turn the raw accounting extract into a firm-year panel rather than a collection of filings. I date each observation by the fiscal closing date and keep one record per BvD firm-year. When more than one filing is available in the same year, the code gives priority to unconsolidated accounts before consolidated accounts, following the order U1, U2, C1, and C2. This matters because duplicate firm-years are often generated by consolidated and unconsolidated filings for the same legal entity. If the primary NACE code is unavailable, I use the secondary NACE code to reduce sector-code attrition. I then map firms from NACE Rev. 2 to 4-digit ISIC Rev. 4 sectors using the NACE–ISIC concordance used throughout the construction scripts.

Nominal ORBIS/AMADEUS accounting values are combined with TURKSTAT producer price indices before the main variables are constructed. The deflator is matched first at the 3-digit NACE level and, where needed, at the 2-digit NACE level. Observations without a usable manufacturing sector or without a matched PPI are dropped. The resulting raw Turkish man-

ufacturing panel used in the thesis covers 15,957 firms and 78,921 firm-year observations from 2009 to 2023. The panel is unbalanced: firms can enter the observed sample, disappear, or report intermittently. I do not impute missing accounting years. For this reason, the thesis interprets disappearance from ORBIS/AMADEUS as observed sample turnover rather than validated legal closure or economic exit.

The firm-level outcome construction follows the same panel logic. Revenue, costs of goods sold, tangible fixed assets, intangible fixed assets, and domestic sales are deflated by the matched PPI. Domestic sales are calculated as operating revenue minus export revenue when export revenue is available. Sector-year domestic market shares are then computed within 4-digit ISIC sectors, and these shares are used to build concentration measures such as CR10, CR20, the domestic HHI, and the firm-count variables used in the heterogeneity and decomposition exercises. Firm age is measured as years since the first observed year in the ORBIS/AMADEUS panel, so it is a panel-age measure rather than a legal incorporation-age measure. Other controls and mechanisms are built from lagged accounting ratios, including firm size, leverage, liquidity, current ratio, working-capital intensity, interest burden, and cash buffers.

The markup measure is revenue-based. The baseline construction uses deflated operating revenue as output revenue, costs of goods sold as the flexible input, and tangible fixed assets as the state input. In the construction scripts, a translog revenue production function is estimated using log revenue, log COGS, log tangible fixed assets, their squares and interaction, and sector and year controls. The implied output elasticity of the flexible input is then combined with the COGS revenue share to obtain firm-year markups. The main dependent variable is the annual change in log markup, so the estimation sample is smaller than the raw firm panel: a firm-year must have positive revenue, COGS, and capital, a valid markup, consecutive observations for first differences, and the lagged controls required by the specification. After these restrictions and the analysis-stage trimming of extreme observations, the main firm-level specifications use about 31,000 firm-year observations. The shift-share treatment design contains 609 effective sector-year cells.

Figure 8 documents the evolution of the number of observed firms over time. The decline in coverage in later years is one reason why the main text treats ORBIS/AMADEUS disappearance cautiously and why the baseline regressions control for pre-period labor intensity interacted with the post-2016 period. Figure 9 summarizes the distribution of sector size in the manufacturing sample and illustrates the granularity behind the shift-share design.

CEPII BACI trade data. CEPII BACI HS96 files provide bilateral trade flows at the HS 6-digit level. In the import-penetration construction, I keep Chinese exports to Turkey, using China as exporter code 156 and Turkey as importer code 792, and aggregate product flows to ISIC Rev. 4

manufacturing sectors. The baseline treatment is

$$IP_{jt} = \frac{M_{jt}^{CHN \rightarrow TUR}}{\text{Apparent consumption}_{jt}},$$

and the regressor used in the firm-level specifications is the annual change in this sectoral import-penetration rate. The BACI files are also used for the shift-share design. The exposure share is based on China's 2009 exports to Turkey by sector relative to sectoral absorption. The shift component is the change in China's export performance in other high-income destinations, including the United States, Japan, South Korea, Australia, and EU-15 economies. A separate BACI construction over 2000–2004 is used in the mechanism section to measure China's pre-period share in Turkey's upstream imports.

UNIDO sector data. UNIDO ISDB/INDSTAT4 files provide sectoral manufacturing aggregates at the 4-digit ISIC level. I use Turkish sectoral output and apparent consumption to construct the denominator of import penetration and the 2009 exposure shares. The construction scripts import Turkish output from the UNIDO file, harmonize sector codes to ISIC Rev. 4, and use the ISDB apparent-consumption series directly when computing import penetration for 2011–2019. A high-income import file is also used to normalize the high-income-destination shift in the output-competition instrument.

WIOD input-output data. The input-supply shifter is built from the Turkish national input-output table in WIOD. The main construction uses the 2003 table. I keep domestic and imported intermediate-flow rows, drop final-demand and total rows, reshape the table into input-sector by using-sector cells, and compute

$$w_{jk} = \frac{\text{intermediate purchases by sector } j \text{ from sector } k}{\text{total intermediate purchases by sector } j}.$$

Manufacturing using sectors are retained, while the denominator includes all intermediate inputs rather than only manufacturing inputs. The output-side sector shock is then propagated through this fixed input-output matrix to obtain the input-supply control $Z_{jt}^{\text{Input}} = \sum_k w_{jk} Z_{kt}^{\text{Output}}$. For the imported-input-dependence mechanism, I also use 2000–2004 WIOD weights and combine them with BACI-based Chinese import shares in upstream sectors.

TURKSTAT data. TURKSTAT data enter in two places. First, the monthly sectoral Producer Price Index is converted into annual sectoral PPI series and used to deflate ORBIS/AMADEUS accounting variables. The construction uses 3-digit NACE deflators where available and falls

back to 2-digit deflators otherwise. Second, TURKSTAT Annual Industry and Service Statistics extracts on personnel costs and value added at factor cost are used to construct sectoral labor intensity. I compute the ratio of personnel costs to value added, average it over the pre-2016 period, fill missing 4-digit values with 3-digit or 2-digit sector means when necessary, and map the resulting measure to ISIC Rev. 4. This time-invariant labor-intensity control is interacted with a post-2016 dummy in the baseline specifications.

HS96–ISIC Rev. 4 concordance.

BACI reports products in the 1996 Harmonized System revision at the 6-digit product level. In the `concordance` package terminology this is HS1, so the construction scripts use the HS96-to-ISIC4 call

```
concord_hs_isic(sourcevar = hs_vec, origin = "HS1", destination =
  "ISIC4", dest.digit = 4, all = TRUE)
```

rather than the package’s HS6 option, which refers to the 2022 HS revision. The underlying package supports HS revisions, ISIC Rev. 2, Rev. 3, Rev. 3.1, and Rev. 4, and reports all destination matches and occurrence-share weights when `all = TRUE` (Liao et al., 2020). Its HS–ISIC tables are based on the World Bank WITS product-concordance files (World Integrated Trade Solution, 2026).

The HS96-to-ISIC4 bridge is many-to-many. The package first uses the HS96-to-ISIC Rev. 3 dictionary and then links ISIC Rev. 3 to ISIC Rev. 4; after retaining distinct matched pairs, it counts how often each 4-digit ISIC destination occurs for a given HS product and converts these counts into shares. I keep all matched ISIC4 sectors and their shares, rather than selecting only the modal sector. For each BACI observation, trade value is therefore split across matched industries as $X_{hjt} = s_{hj}X_{ht}$, where h is the HS96 6-digit product, j is the ISIC Rev. 4 sector, and $\sum_j s_{hj} = 1$ over the matched sectors for product h . The allocated values are then collapsed to sector-year totals before constructing import penetration, the shift-share exposure, and the Chinese product-granularity robustness measure.

The practical implication is that the sectoral China-exposure variables are not built by forcing each HS6 product into a single industry. They preserve the concordance’s fractional mapping whenever a product spans multiple ISIC4 industries. The main limitation is that the shares are occurrence shares from the concordance crosswalk, not Turkish-use, production, or import-value shares; they therefore reduce arbitrary modal assignment but cannot recover the actual industry use of every HS product in Turkey.

Concordances and replication note. Two concordance layers are used repeatedly. The OR-BIS/AMADEUS firm data are mapped from NACE Rev. 2 to ISIC Rev. 4 using the NACE–ISIC

file in the raw data folder. BACI product flows are mapped from HS96 6-digit products to ISIC Rev. 4 industries using the HS–ISIC concordance generated by the R scripts in `code/01_construct/`. A fully public replication package cannot be distributed because the main firm-level source, ORBIS/AMADEUS, is proprietary. Replication of the full empirical analysis therefore requires licensed access to ORBIS/AMADEUS and compliance with the terms of use of the other underlying datasets, including BACI, WIOD, UNIDO, and TURKSTAT. Within the project archive, the construction scripts are organized in `code/01_construct/` and `code/02_analysis/`, with derived tables and figures stored in `output/tables/` and `output/figures/`. Public sharing is feasible for code, concordances, and non-restricted sectoral aggregates, but not for the proprietary firm-level extracts themselves.

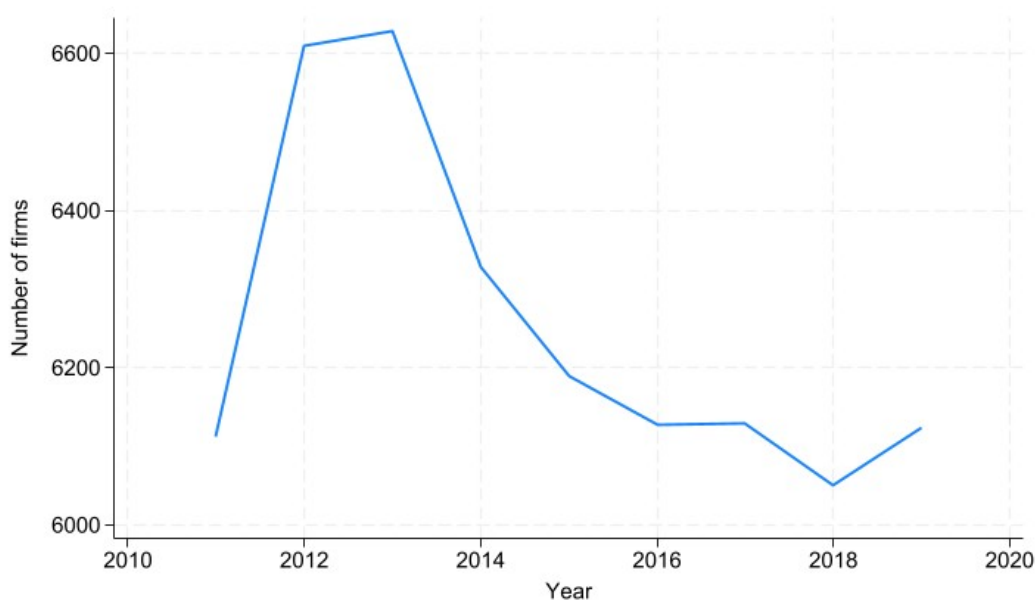


Figure 8: Observed ORBIS/AMADEUS manufacturing firms by year. The horizontal axis is year and the vertical axis is the number of observed firms.

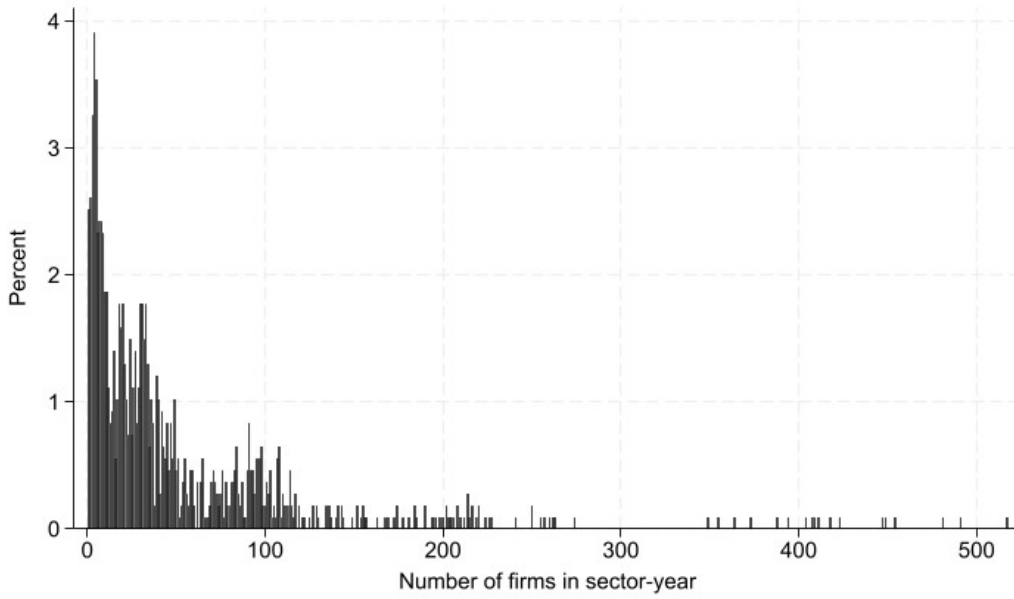


Figure 9: Distribution of sector-year cell sizes. The horizontal axis is the number of firms in an ISIC4 sector-year cell and the vertical axis is percent.

B.2 Empirical Strategy and Main Results

B.2.1 Grouped-IV quantile estimates

Tables 9 and 10 report the numerical grouped-IV quantile estimates underlying Figure 7. The outcome in each column is the corresponding sector-year quantile of firm-level markup adjustment, and the treatment remains instrumented sectoral import-penetration growth.

Table 9: Grouped IV quantiles of markup adjustment (Panel A: p5–p50)

	q5	q10	q15	q20	q25	q30	q35	q40	q45	q50
Import penetration change (ΔIP)	-0.531 (0.506)	0.154 (0.429)	0.245 (0.207)	-0.00378 (0.217)	0.0130 (0.185)	-0.0290 (0.160)	-0.0971 (0.160)	-0.158 (0.154)	-0.223 (0.156)	-0.195 (0.141)
Input-supply shock (Z^{Input})	5.836* (3.339)	3.821 (2.725)	0.800 (1.229)	-0.0735 (0.694)	-0.396 (0.648)	-0.334 (0.628)	0.0461 (0.550)	0.0166 (0.526)	0.0383 (0.506)	0.0150 (0.489)
Sector-year observations	609	609	609	609	609	609	609	609	609	609

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Grouped IV quantiles of markup adjustment (Panel B: p55–p95)

	q55	q60	q65	q70	q75	q80	q85	q90	q95
Import penetration change (ΔIP)	-0.182 (0.148)	-0.219 (0.147)	-0.235 (0.158)	-0.108 (0.148)	-0.246 (0.160)	-0.848*** (0.313)	-1.423** (0.597)	-1.585*** (0.550)	-1.462* (0.748)
Input-supply shock (Z^{Input})	-0.176 (0.504)	0.0432 (0.559)	0.141 (0.592)	1.474 (1.560)	1.476 (1.563)	1.465 (1.757)	1.782 (2.125)	1.709 (2.215)	6.899** (3.199)
Sector-year observations	609	609	609	609	609	609	609	609	609

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The tables show that the lower and middle quantiles are close to zero and mostly imprecise, while the upper quantiles become large, negative, and statistically significant from around the 80th percentile onward. This pattern is the quantitative basis for the claim that Chinese import competition compresses the upper tail of markup adjustments rather than shifting the entire distribution uniformly downward. It also explains why the average firm-level effect can be significant while the sector-level aggregate effect is muted: the main action is concentrated among firms with unusually high positive markup growth.

B.3 Threats to Identification and Robustness

B.3.1 Control-set sensitivity

Table 11 reports the specification checks behind the robustness discussion in the main text. The table varies the control set, excludes the input-supply control, and adds a quadratic term. The output-competition coefficient remains negative across the linear specifications, so the baseline result is not driven by one narrow control choice.

Table 11: Specification diagnostics: functional form and control-set sensitivity

Spec	Output-shock effect (ΔIP)	Input-shock control (Z^{Input})	Quadratic term (ΔIP^2)	Obs.	Wild p-value	ID	Control-set design
S1	-0.462** (0.226)	0.517 (0.470)	–	30551	0.026	(1)	Baseline
S2	-0.101 (0.485)	0.578 (0.391)	7.783 (6.489)	30551	.	(2)	Quadratic
S3	-0.462** (0.226)	0.518 (0.471)	–	30551	.	(3)	No age
S4	-0.462** (0.227)	0.519 (0.471)	–	30551	.	(4)	No age, size
S5	-0.461** (0.226)	0.521 (0.470)	–	30551	.	(5)	No age, size, exporter
S6	-0.441* (0.225)	0.497 (0.463)	–	30566	.	(6)	No age, size, exporter, liquidity
S7	-0.442* (0.225)	0.495 (0.463)	–	30566	.	(7)	No age, size, exporter, liquidity, LSxPost
S8	-0.409* (0.226)	–	–	30655	.	(8)	No input shock
S9	-0.396* (0.230)	–	–	31496	.	(9)	No controls

Cluster-robust standard errors (ISIC4) in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3.2 Identification: exposure, pre-trends, and influential sectors

This subsection reports identification and sample-composition checks on the baseline estimate's empirical support. The first documents the sectors carrying the largest Rotemberg weights and asks how much the IV estimate depends on them. The second adds

sector-level demand controls and pre-trend balance tests. The remaining checks vary the treatment of extreme markup changes and examine whether the result survives restrictions to longer-lived or size-filtered firms.

Table 12: Top-weighted sectors: Rotemberg weights and sectoral economic size

ISIC4	Sector Name	Signed Rot. Weight	Abs. Rot. Weight	2009 Output Share (%)	Cumul. Output Share (%)	China IP 2009
2610	Electronic components and boards	0.489440	0.489440	0.098	0.098	0.6090
1629	Other wood products;articles of cork,str	0.173203	0.173203	0.047	0.144	0.4756
3290	Other manufacturing n.e.c.	0.141033	0.141033	0.217	0.362	0.4998
2513	Steam generators, excl. hot water boiler	0.081717	0.081717	0.046	0.408	0.1150
2431	Casting of iron and steel	0.053969	0.053969	0.395	0.802	0.1628

Notes: Rotemberg (signed and absolute) weights are normalized sector contributions to the first-stage variance. Output shares are pre-period (2009) sector value added relative to total manufacturing. China IP 2009 is the baseline exposure share to Chinese imports.

Table 12 reports the five sectors with the largest absolute Rotemberg weights and compares them with their pre-period economic size. The largest weights are not large output-share sectors; they are small sectors with high initial China exposure. This supports the local interpretation emphasized in the main text.

Table 13: IV robustness: demand-control checks

	(1) $\Delta \log \mu$	(2) $\Delta \log \mu$	(3) $\Delta \log \mu$	(4) $\Delta \log \mu$
Import penetration change	-0.462** (0.226)	-0.458** (0.226)	-0.462** (0.224)	-0.459** (0.227)
Input-supply shock	0.517 (0.470)	0.565 (0.478)	0.534 (0.471)	0.556 (0.476)
High-income demand growth		0.029 (0.022)		0.018 (0.030)
Turkish export-demand growth			0.019 (0.012)	0.010 (0.014)
Observations	30,551	30,542	30,542	30,542

Standard errors in parentheses, clustered by ISIC4 sector.

All regressions instrument import penetration change with the output-competition IV.

Controls include lagged firm controls, the input-supply shock, labor-share-by-post-2016 interaction, ISIC4 FE, and year FE.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13 adds two demand-side controls to the baseline IV specification. The high-income global-demand control absorbs sectoral demand shifts in the same foreign markets used to construct the China shifter, while the Turkish export-demand control absorbs shocks to external demand faced by Turkish producers. The import-penetration coefficient remains close to -0.46 in all columns, so the baseline result is not explained by these demand trends.

Table 14: Pre-trend balance checks: do future China shocks predict pre-period outcomes?

	(1) $\Delta \log \mu$	(2) $\Delta \log \text{revenue}$	(3) $\Delta \log \text{HHI}$
Future output-competition shock	-0.062 (0.631)	0.152 (1.967)	0.710 (12.088)
Future input-supply shock	-2.839 (3.011)	-15.610*** (3.421)	37.522* (20.327)
Observations	275	279	185
R-squared	0.032	0.056	0.007

Standard errors in parentheses, clustered by ISIC4 sector.

Each column is a sector-year pre-period outcome regression.

The main coefficient tests whether sectors more exposed to later China shocks already had differential pre-period trends. Regressions include year fixed effects.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14 tests whether future China shocks predict pre-period sectoral outcomes. The future output-side shock is small and statistically insignificant for pre-period markup growth, revenue growth, and domestic concentration. This supports the timing assumption behind the shift-share design, while the significant coefficients on the future input shock are treated as a warning to keep the input shifter as a control rather than as the excluded instrument in the main markup equation.

Table 15: Robustness to excluding sectors with largest Rotemberg weights

	Baseline	Excl. top 5	Excl. top 10	Excl. top 10%
Panel A. Second stage				
Import penetration change (ΔIP)	-0.462** (0.226)	-0.316 (1.889)	0.246 (2.285)	2.008 (4.575)
Input-supply shock (Z^{Input})	0.517 (0.470)	0.336 (0.625)	0.394 (0.553)	0.444 (0.594)
Observations	30551	29393	28512	27669
KP rk Wald F	24.81	3.178	8.129	4.759
Panel B. First stage				
Output-competition IV (Z^{Output})	0.892*** (0.179)	0.324* (0.182)	0.382*** (0.134)	0.241** (0.110)
Input-supply shock (Z^{Input})	0.163 (0.491)	-0.242 (0.198)	-0.150 (0.148)	-0.055 (0.122)
Observations	30551	29393	28512	27669

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 15 shows that the first stage is concentrated. Removing the five most influential sectors lowers the Kleibergen–Paap rk Wald F -statistic sharply, and broader exclusions make the second-stage estimates unstable. This does not contradict the baseline result; it clarifies its scope. The main estimate is credible for the exposed sector-year cells that carry identifying weight, but it should not be presented as a uniformly precise average for every Turkish manufacturing sector.

B.3.3 Residualized Output-IV Results

The next four appendix tables report the robustness exercise in which the output-competition instrument is residualized on the input-supply shock before re-estimation. This directly addresses the concern that the excluded instrument is partly proxying for upstream China exposure. The residualization is done at the sector-year level, so the remaining variation is the component of

the output shock orthogonal to the input-supply shifter.

Table 16: First-stage comparison: original vs residualized output IV

	(1) Main IV	(2) Residualized IV
Output-competition IV (Z^{Output})	0.892*** (0.179)	–
Residualized output IV ($\tilde{Z}^{\text{Output}}$)	–	0.892*** (0.179)
Input-supply shock (Z^{Input})	0.163 (0.491)	0.524 (0.490)
Observations	30551	30551
R^2	0.149	0.149
Instrument	Original output IV	Residualized output IV

Standard errors in parentheses.

Residualized IV is the component of Z^{Output} orthogonal to Z^{Input} .

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 16 reports the first stage after residualizing the output-side shifter on the input-supply shifter. The residualized instrument remains strong, so the output-competition variation is not exhausted by upstream exposure.

Table 17 re-estimates the baseline markup equation with the residualized output instrument. The coefficient is unchanged at about -0.462 , supporting the main text’s claim that the incumbent markup result comes from output competition.

Table 18 repeats the residualized-IV exercise across fixed-effect and control specifications. The sign and magnitude remain stable across these variants, reinforcing the robustness interpretation.

Table 19 applies the residualized-IV design to auxiliary outcomes. The signs continue to support a domestic-demand adjustment margin rather than an input-channel confound.

The exercise leaves the main interpretation unchanged. The first stage remains strong, the markup coefficient is numerically identical to the baseline estimate, and the signs of the auxiliary outcomes are preserved. This stability is important because the output and input variables are mechanically related through China’s export growth; after removing their linear overlap, the pro-competitive markup result still comes from the output-competition component.

Table 17: Markup response: original vs residualized output IV

	(1) Main IV	(2) Residualized IV
Import penetration change (ΔIP)	-0.462** (0.226)	-0.462** (0.226)
Input-supply shock (Z^{Input})	0.517 (0.470)	0.517 (0.470)
Observations	30551	30551
KP rk Wald F	24.81	24.81
Wild-cluster bootstrap p -value	—	—
Instrument	Original output IV	Residualized output IV

Standard errors in parentheses.

Wild-cluster bootstrap p -values are not reported for this auxiliary appendix table.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 18: Residualized output IV: functional-form and control sensitivity

Spec	Import-penetration effect (ΔIP)	Input-supply shock (Z^{Input})	Quadratic term (ΔIP^2)	Obs.	Control-set design
S1	-0.462** (0.226)	0.517 (0.470)	—	30551	Baseline
S2	-0.193 (0.510)	0.562 (0.398)	5.799 (6.493)	30551	Quadratic
S3	-0.462** (0.226)	0.518 (0.471)	—	30551	No age
S4	-0.462** (0.227)	0.519 (0.471)	—	30551	No age, size
S5	-0.461** (0.226)	0.521 (0.470)	—	30551	No age, size, exporter
S6	-0.441* (0.225)	0.497 (0.463)	—	30566	No age, size, exporter, liquidity
S7	-0.442* (0.225)	0.495 (0.463)	—	30566	No age, size, exporter, liquidity, LS×Post
S8	-0.459** (0.226)	—	—	30566	No input shock
S9	-0.439* (0.231)	—	—	31403	No controls

Cluster-robust standard errors (ISIC4) in parentheses.

Wild-cluster bootstrap p -values are not reported for this auxiliary appendix table.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 19: Main-text outcome validation with residualized output IV

Outcome	Parameter	Main IV	Residualized IV	Obs.	KP rk Wald F
Panel A. Import-penetration effect					
Markup adjustment	Import penetration change (ΔIP)	-0.462** (0.226)	-0.462** (0.226)	30551	24.81
Revenue growth	Import penetration change (ΔIP)	-1.639*** (0.537)	-1.639*** (0.537)	24345	26.97
Domestic-sales growth	Import penetration change (ΔIP)	-1.636** (0.658)	-1.636** (0.658)	30551	24.81
Export growth	Import penetration change (ΔIP)	-1.123 (2.452)	-1.123 (2.452)	19327	32.57
Panel B. Input-supply shock					
Markup adjustment	Input-supply shock (Z^{Input})	0.517 (0.470)	0.517 (0.470)	30551	24.81
Revenue growth	Input-supply shock (Z^{Input})	2.746* (1.653)	2.746* (1.653)	24345	26.97
Domestic-sales growth	Input-supply shock (Z^{Input})	0.813 (1.942)	0.813 (1.942)	30551	24.81
Export growth	Input-supply shock (Z^{Input})	-7.200 (5.215)	-7.200 (5.215)	19327	32.57

Standard errors in parentheses.

The main and residualized IV columns are numerically identical for all four outcomes.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3.4 Markup measurement and sample selection

The following tables support the robustness discussion on markup measurement and observed panel composition. The alternative Kirov et al. estimator asks whether the result survives a more demanding markup construction, while the trimming and long-lived-firm tables ask whether the estimate is an outlier artifact or depends on selective sample restrictions.

B.3.5 Alternative markup estimation

KMT is designed to treat omitted-price bias and allow markup heterogeneity across firms and over time. KMT methodology consists of two steps: first, recover a revenue elasticity, which is contaminated by the markup, from the flexible-input share equation; second, use a Markov process for revenue productivity to back out the true physical output elasticity and thus the markup level. The KMT approach is more robust to misspecification of the production function and allows for more flexible markup dynamics, but it also relies on stronger assumptions about the evolution of productivity and the structure of demand.

Revenue production function: In the implemented robustness, I use a Cobb-Douglas baseline:

$$r_{it} = \beta_x x_{it} + \beta_k k_{it} + \nu_{it} + \varepsilon_{it}$$

where ν_{it} is the revenue productivity shock that follows a first-order Markov process: $\nu_{it} = \rho \nu_{it-1} + \eta_{it}$, and ε_{it} is an i.i.d. shock. The key difference from the standard production function estimation is that the dependent variable is revenue rather than physical output, which means that the estimated β_x captures both the physical output elasticity with respect to the flexible input and the markup.

Revenue productivity Markov process: To identify the physical output elasticity, KMT exploits the first-order Markov structure of revenue productivity as follows:

$$\nu_{it} = \rho g(\nu_{it-1}) + \eta_{it}$$

where $g(\cdot)$ is a non-parametric function that captures the potentially non-linear evolution of revenue productivity. I approximate $g(\cdot)$ using a third-order polynomial in lagged revenue productivity.

Markup control function: Let markups be a function of inputs, time and firm fixed effects, and a vector of both firm- and sector-level time-varying parameters $\mathbb{D}_{i(j)t-1}$ that shift markups but are not included in the production function estimation. These parameters include lagged share of sales, lagged exporter status, and lagged HHI at the ISIC4 level, which are all potential confounders of the markup estimation.

Share equation: I then regress the log flexible-input cost share in revenue ($s_{it}^x \equiv cx_{it} - r_{it}$) on log inputs and markup shifters:

$$s_{it}^x = \ln \beta_x - \mu_{it} + b - \varepsilon_{it}$$

where $\ln(\beta_x)$ is the technological component, which captures the effect of the flexible input on revenue. μ_{it} is the markup component, which captures the effect of markups on the cost share. Higher markups lower the cost share, so it enters with a negative sign. $b - \varepsilon_{it}$ is the error term, which states that even after firms choose inputs, realized cost shares may differ from the predicted cost shares due to measurement error or other unobserved factors.

In my analysis, the share equation is estimated as follows:

$$\begin{aligned} \ln(\text{COGS}_{it}) - \ln(R_{it}) = & \ln \beta_{\text{COGS}} - (\phi_0 + \phi_x x_{it} + \phi_k k_{it} + \phi_1 \text{share_sales}_{it-1} \\ & + \phi_2 \text{exporter_status}_{it-1} + \phi_3 \text{HHI}_{jt-1} + \tau_t + \delta_j + \lambda_i) + b - \varepsilon_{it}. \end{aligned}$$

Final markup estimator: After recovering $\hat{\theta}^{\text{COGS}} = \hat{\beta}_{\text{COGS}}$, I compute:

$$\hat{\mu}_{it}^{\text{KMT}} = \frac{\hat{\theta}^{\text{COGS}}}{\text{COGS}_{it}/R_{it}}.$$

I then define the KMT markup adjustment as $\Delta \log \hat{\mu}_{it}^{\text{KMT}} = \log \hat{\mu}_{it}^{\text{KMT}} - \log \hat{\mu}_{i,t-1}^{\text{KMT}}$, which is the dependent variable in the robustness regression.

Table 20: Robustness using Kirov et al. (2026) markup estimates

	(1)	(2)
	Dep. var.: $\Delta \ln \mu^{\text{KMT}}$	Dep. var.: $\ln \mu^{\text{KMT}}$
Import penetration change (ΔIP)	-0.446 (0.518)	-0.143 (0.397)
Input-supply control (Z^{Input})	1.028 (0.955)	0.937 (0.684)
Mean pre-period labor share	0.101 (0.089)	-0.146 (0.121)
Post-2016 indicator	0.012 (0.018)	0.032 (0.024)
Post-2016 $\times$ mean labor share	-0.007 (0.033)	-0.076* (0.044)
Observations	23039	29511
α_M	0.940	0.940
α_K	0.060	0.060
b	0.024	0.024

Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 21: Trimming/winsorization

	(1)	(2)	(3)
	$\Delta \log \mu$	$\Delta \log \mu$ (winsorized 5–95)	$\Delta \log \mu$ (winsorized 0.5–99.5)
Import penetration change (ΔIP)	-0.462** (0.226)	-0.323** (0.144)	-0.524** (0.231)
Input-supply shock (Z^{Input})	0.517 (0.470)	0.133 (0.255)	0.332 (0.359)
Observations	30551	30551	30551

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 21 shows that the negative markup response is not an outlier artifact. The point estimate remains negative and statistically significant under both tighter and looser winsorization thresholds. The magnitude changes modestly, as expected when the dependent variable is a first difference of an accounting-based markup, but the sign and interpretation are stable.

Table 22 restricts the sample in ways that progressively remove short-lived, intermittently

Table 22: The case of long-lived firms and exclusion of micro and top firms by size

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	$\Delta \log \mu$	$\Delta \log \mu$	$\Delta \log \mu$	$\Delta \log \mu$	$\Delta \log \mu$	$\Delta \log \mu$	$\Delta \log \mu$	$\Delta \log \mu$
Import penetration change (ΔIP)	0.0462 (0.138)	0.0348 (0.111)	-0.194 (0.180)	-0.301 (0.217)	-0.344 (0.248)	-0.540 (0.335)	-0.525*** (0.179)	-0.448** (0.225)
Input-supply shock (Z^{Input})	-0.0171 (0.581)	-0.0920 (0.373)	0.0144 (0.544)	0.0411 (0.472)	0.145 (0.427)	0.0510 (0.377)	0.225 (0.425)	0.477 (0.470)
Lagged leverage	0.0123** (0.00597)	0.0147* (0.00746)	0.0215*** (0.00478)	0.0195*** (0.00414)	0.0158*** (0.00432)	0.0124*** (0.00409)	0.0112** (0.00447)	0.00740* (0.00395)
Lagged liquidity ratio	-0.000942* (0.000504)	-0.000810 (0.000506)	-0.000584 (0.000411)	-0.000762** (0.000306)	-0.000557** (0.000235)	-0.000356 (0.000495)	-0.000975 (0.00125)	-0.00142 (0.000913)
Lagged log firm size	-0.00269*** (0.000606)	-0.00137* (0.000742)	-0.00155*** (0.000501)	-0.00131*** (0.000403)	-0.000303 (0.000614)	0.000295 (0.000719)	0.0000996 (0.000750)	0.000403 (0.000797)
Lagged firm age	0.0000308 (0.000318)	-0.000590 (0.000561)	-0.000511 (0.000376)	-0.000318 (0.000266)	-0.000185 (0.000258)	-0.000244 (0.000248)	-0.0000220 (0.000252)	-0.000127 (0.000302)
Lagged exporter status	0.00214 (0.00273)	-0.00271 (0.00398)	-0.00410 (0.00300)	-0.00145 (0.00256)	-0.000146 (0.00218)	0.00104 (0.00176)	0.00191 (0.00168)	0.00300 (0.00187)
Observations	4136	5473	10531	13661	17501	23138	27492	30249

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

reporting, or size-extreme firms. The coefficient attenuates in the most restrictive long-lived samples and becomes more stable as the sample approaches the baseline coverage. This pattern is informative: conditioning on firms with long reporting histories removes precisely the panel-composition margin that import competition may affect. The full unbalanced panel is therefore the preferred specification for estimating within-incumbent markup discipline among observed firms.

B.3.6 Distributional design

The following figures and Monte Carlo tables support the grouped-IV quantile design in Section 5.4. They vary the quantile construction and show why the specifications control flexibly for sector-year cell size, especially in the upper tail.

B.3.7 Grouped-IV Quantile Robustness Figures

Figures 10–12 examine whether the upper-tail quantile result depends on arbitrary implementation choices. The baseline distributional evidence uses deciles of markup adjustment within sector-year cells. These figures vary the binning, the ranking variable, and the minimum cell-size requirement.

Figure 10 shows that using quintiles instead of deciles preserves the main shape of the response. The estimated effects remain small through the lower and middle parts of the distribution and become more negative toward the upper tail.

Figure 11 ranks firms by lagged markup levels rather than current markup adjustments. This is a demanding check because it asks whether firms that enter the period with high measured market power show stronger discipline. The estimates remain most negative near the upper

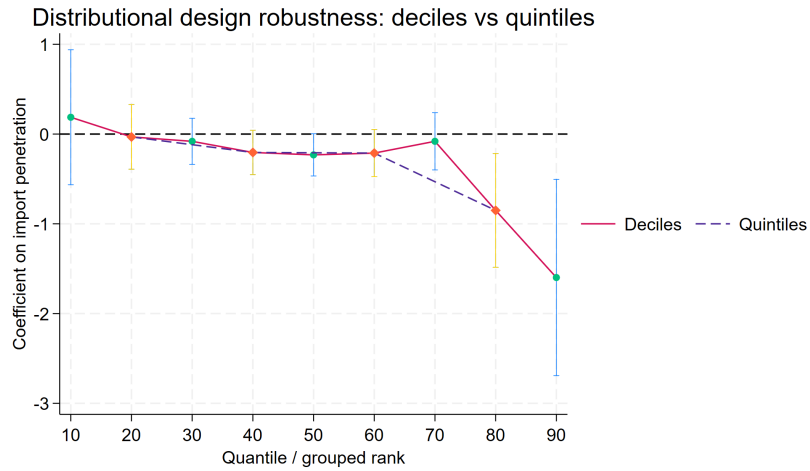


Figure 10: Distributional design robustness: deciles versus quintiles

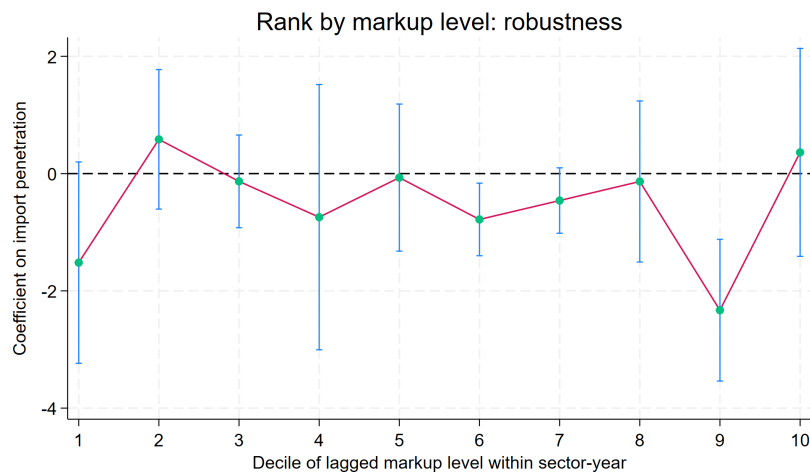


Figure 11: Distributional design robustness: ranking by markup level

ranks, reinforcing the interpretation that the result is connected to market-power states rather than to a mechanical feature of the adjustment distribution.

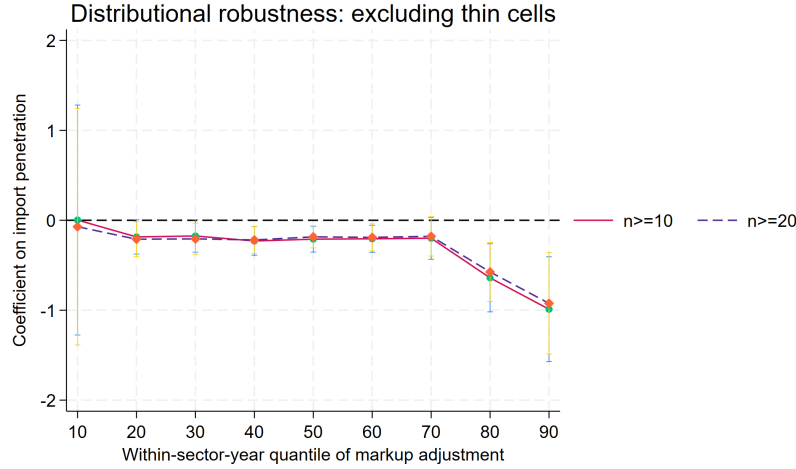


Figure 12: Distributional design robustness: excluding thin cells

Figure 12 excludes thin sector-year cells where empirical quantiles are most sensitive to small-sample order-statistic noise. The upper-tail pattern remains visible after this restriction, suggesting that the baseline quantile profile is not driven by a few small cells.

B.3.8 Monte Carlo Diagnostics

The next tables report the finite-sample diagnostics behind the selection diagnostic and grouped-IV quantile correction. The first table uses simulated firm panels with endogenous observation. The second and third tables use simulated sector-year cells to show how the grouped-quantile correction behaves when cell size is correlated with instrument variation.

B.3.9 Chinese Product Granularity

The next appendix tables examine whether the baseline and distributional estimates are explained by product-level concentration in Chinese supply to Turkey. Because firm-level Chinese exporter data are unavailable, the proxy uses BACI HS6 products mapped to ISIC4 sectors and computes the product HHI of Chinese imports within each sector-year. The preferred specifications use lagged Chinese product concentration as a predetermined control or interaction term.

Table 26 adds lagged Chinese product concentration to the mean IV specification. The baseline import-penetration coefficient remains negative, so product granularity does not explain away the incumbent markup result.

Table 23: Monte Carlo diagnostic for the selection control function

True effect	Specification	Mean beta	Bias	Mean SE	Coverage	Rejection rate
$\beta_0 = 0$	Cubic polynomial	-0.0001	-0.0001	0.0118	0.946	0.054
	Fifth-order polynomial	-0.0001	-0.0001	0.0118	0.946	0.054
	Spline, 6 bases	-0.0001	-0.0001	0.0118	0.948	0.052
	Uncorrected	0.1134	0.1134	0.0133	0.000	1.000
$\beta_0 = 0.5$	Cubic polynomial	0.4999	-0.0001	0.0118	0.946	1.000
	Fifth-order polynomial	0.4999	-0.0001	0.0118	0.946	1.000
	Spline, 6 bases	0.4999	-0.0001	0.0118	0.948	1.000
	Uncorrected	0.6134	0.1134	0.0133	0.000	1.000

Notes: The table reports Monte Carlo results for the firm-level selection control function with $\theta = 0.5$. Coverage is the share of 95 percent confidence intervals containing the true coefficient. For $\beta_0 = 0$, the rejection rate is the rejection rate of the zero null. For $\beta_0 = 0.5$, it is the rejection rate against zero. Each design uses 1,000 replications. The average selection rate is approximately 0.491.

Table 24: Monte Carlo diagnostic for grouped-IV quantile order-statistic bias

Quantile	Mean coefficient		Coverage		5% rejection rate	
	Uncorrected	Corrected	Uncorrected	Corrected	Uncorrected	Corrected
10	0.0104	0.0126	0.666	0.770	0.334	0.230
20	0.0070	0.0063	0.756	0.880	0.244	0.120
30	0.0061	0.0055	0.758	0.888	0.242	0.112
40	0.0057	0.0045	0.796	0.902	0.204	0.098
50	0.0060	0.0031	0.768	0.920	0.232	0.080
60	0.0062	0.0043	0.752	0.904	0.248	0.096
70	0.0070	0.0050	0.702	0.886	0.298	0.114
80	0.0092	0.0053	0.564	0.888	0.436	0.112
90	0.0148	0.0082	0.306	0.848	0.694	0.152

Notes: The table reports a null Monte Carlo with true grouped-IV quantile effects equal to zero. The corrected specification includes a cubic polynomial in sector-year cell size. Coverage is the share of 95 percent confidence intervals containing zero. Rejection is the share of tests rejecting the zero null at the 5 percent level. The simulation uses 500 replications.

Table 25: Upper-tail spline-bootstrap robustness for grouped-IV quantiles

Method	Quantile	Mean beta, corrected	Rejection, uncorrected	Rejection, corrected
Polynomial control	80	0.0058	0.436	0.120
Polynomial control	90	0.0089	0.692	0.165
Spline + bootstrap	80	0.0031	0.450	0.000
Spline + bootstrap	90	0.0025	0.750	0.050

Notes: The table reports a targeted upper-tail null Monte Carlo. The spline-bootstrap correction uses a six-basis spline in sector-year cell size with bootstrap inference. Because the procedure is computationally costly, it is implemented only for Q80 and Q90.

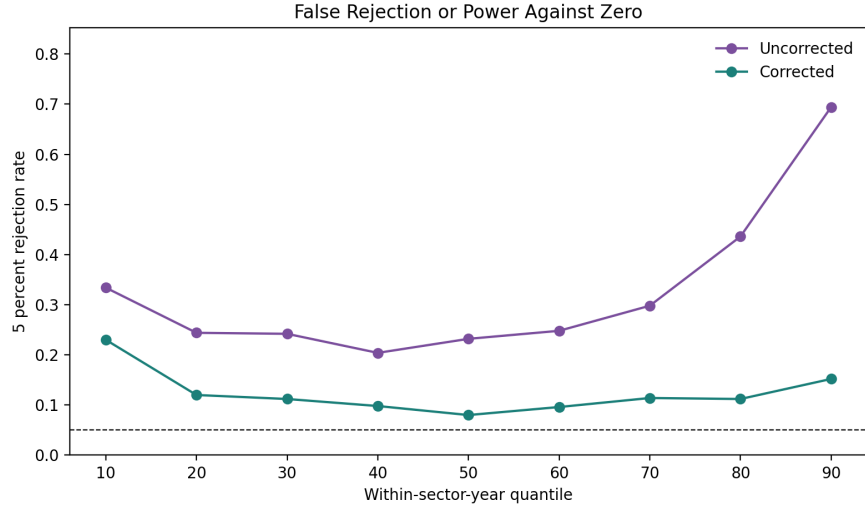


Figure 13: False rejection rates in the grouped-IV quantile Monte Carlo
Notes: The true effect is zero at every quantile. The dashed horizontal line denotes the nominal 5 percent rejection rate. The corrected estimator includes a cubic polynomial in sector-year cell size.

Table 26: Chinese product granularity and incumbent markup adjustment

	(1) Baseline	(2) + China product HHI	(3) China product HHI $\times$ IP	(4) + China IP-HHI	(5) China IP-HHI $\times$ IP	(6) Domestic + foreign granularity
Chinese import-penetration growth	-0.463** (0.226)	-0.464** (0.228)	-0.549*** (0.197)	-0.509* (0.273)	-0.411 (0.717)	-0.620** (0.312)
China IP $\times$ Chinese product HHI			-0.383 (0.265)			-0.484 (0.296)
China IP $\times$ Chinese IP-HHI exposure					-0.0278 (0.147)	
China IP $\times$ domestic HHI						0.0669 (0.215)
Lagged Chinese product HHI, standardized		-0.00105 (0.00294)	-0.00235 (0.00347)			-0.00267 (0.00348)
Lagged Chinese IP-HHI exposure, standardized				-0.00754 (0.00522)	-0.00709 (0.00708)	
Lagged domestic HHI, standardized						-0.000755 (0.00221)
Chinese input-supply exposure	0.531 (0.477)	0.526 (0.480)	0.481 (0.467)	0.445 (0.442)	0.435 (0.426)	0.517 (0.479)
IP effect at Chinese granularity -1 s.d.	.	.	-0.166	.	-0.383	-0.136
IP effect at Chinese granularity mean	-0.463	-0.464	-0.549	-0.509	-0.411	-0.620
IP effect at Chinese granularity $+1$ s.d.	.	.	-0.932	.	-0.439	-1.104
Observations	30269	30269	30269	30269	30269	30269
Kleibergen-Paap rk Wald F	24.79	24.07	12.78	18.70	7.51	0.44
Specification	Baseline	+ China product HHI	China product HHI interaction	+ China IP-HHI	China IP-HHI interaction	Domestic HHI + China product HHI

All specifications include ISIC4 sector fixed effects, year fixed effects, lagged firm controls, the input-supply exposure control, and pre-period size $\times$ post-2016 controls. Chinese import penetration and its interaction terms are instrumented by the corresponding output-IV terms. Heterogeneity variables are standardized using the baseline Chinese-granularity estimation sample. Standard errors are clustered by ISIC4 sector.

Table 27: Relative oligopoly triple interaction

	(1) Triple interaction
Chinese import-penetration growth	-0.945*** (0.350)
China IP $\times$ domestic HHI	0.196 (0.151)
China IP $\times$ Chinese product HHI	-1.118 (0.757)
China IP $\times$ domestic HHI $\times$ Chinese product HHI	0.117 (0.129)
Lagged domestic HHI, standardized	-0.00158 (0.00200)
Lagged Chinese product HHI, standardized	-0.00519 (0.00572)
Chinese input-supply exposure	0.561 (0.522)
Observations	30269
Kleibergen-Paap rk Wald F	1.80
Specification	Triple interaction

This specification estimates whether the China import-competition effect varies jointly with domestic concentration and Chinese product granularity. Chinese import penetration and all interaction terms are instrumented by the corresponding output-IV terms. Standard errors are clustered by ISIC4 sector.

Table 27 interacts import penetration with Chinese product granularity and domestic concentration. The weak interacted first stages make this a diagnostic table, not a new mechanism estimate.

Table 28: Grouped IV quantiles: Chinese product granularity interaction

	(1) Q25	(2) Q50	(3) Q75	(4) Q90
Chinese import-penetration growth	-0.0467 (0.202)	-0.200 (0.160)	-0.133 (0.258)	-1.363** (0.523)
China IP $\times$ Chinese product HHI	0.00505 (0.0770)	-0.0276 (0.0673)	-0.0727 (0.116)	-0.260 (0.229)
Lagged Chinese product HHI, standardized	0.000861 (0.00341)	0.00187 (0.00320)	-0.00454 (0.00590)	0.000307 (0.00660)
Chinese input-supply exposure	-0.203 (0.655)	0.290 (0.408)	1.397 (1.427)	1.319 (2.088)
Sector-year observations	596	596	596	596
Kleibergen-Paap rk Wald F	8.24	8.24	8.24	8.24
Quantile	Q25	Q50	Q75	Q90

The dependent variable is the indicated sector-year quantile of firm-level markup growth. All specifications include ISIC4 sector fixed effects, year fixed effects, sector-year cell-size polynomial controls, averaged lagged firm controls, the input-supply exposure control, and pre-period size $\times$ post-2016 controls. Chinese import penetration and its interaction with Chinese product granularity are instrumented by the corresponding output-IV terms. Standard errors are clustered by ISIC4 sector.

Table 28 adds the foreign product-granularity interaction to the grouped-IV quantile design. The upper-tail compression remains visible, supporting the distributional result in the main text.

Table 29: Grouped IV quantiles: domestic and Chinese granularity

	(1) Q25	(2) Q50	(3) Q75	(4) Q90
Chinese import-penetration growth	1.072 (1.535)	0.716 (1.384)	0.633 (1.212)	-0.929 (1.434)
China IP $\times$ domestic HHI	-1.221 (1.587)	-0.999 (1.403)	-0.831 (1.205)	-0.483 (1.355)
China IP $\times$ Chinese product HHI	2.332 (3.099)	1.878 (2.733)	1.529 (2.323)	0.622 (2.561)
Lagged domestic HHI, standardized	0.0260 (0.0270)	0.0210 (0.0239)	0.0152 (0.0200)	0.0154 (0.0247)
Lagged Chinese product HHI, standardized	0.00771 (0.0126)	0.00748 (0.0109)	0.000212 (0.0113)	0.00283 (0.0122)
Chinese input-supply exposure	-1.352 (1.746)	-0.653 (1.626)	0.581 (1.954)	0.932 (2.409)
Sector-year observations	596	596	596	596
Kleibergen-Paap rk Wald F	0.34	0.34	0.34	0.34
Quantile	Q25	Q50	Q75	Q90

The dependent variable is the indicated sector-year quantile of firm-level markup growth. All specifications include ISIC4 sector fixed effects, year fixed effects, sector-year cell-size polynomial controls, averaged lagged firm controls, the input-supply exposure control, and pre-period size $\times$ post-2016 controls. Chinese import penetration and its interactions with domestic concentration and Chinese product granularity are instrumented by the corresponding output-IV terms. Standard errors are clustered by ISIC4 sector.

Table 29 adds domestic–foreign product-granularity interactions to the grouped-IV quantile design. These specifications are weakly identified and are included only to bound a possible relative-oligopoly interpretation.

The stable baseline columns support the interpretation in the main text: Chinese product

granularity does not explain away the negative incumbent markup response. The distributional robustness is similar. The upper-tail grouped-IV quantile effect remains negative after adding the foreign product-granularity interaction. The domestic–foreign interaction and triple-interaction specifications have weak first stages, so they are included only as diagnostics on a possible relative-oligopoly mechanism.

B.3.10 Synthetic-Control Counterfactual Figures

I implement a synthetic-control exercise for the highly exposed sector ISIC4 2790, with the outcome being the sector inverse markup (Figures 14–17). Following Abadie (2021), the method constructs a weighted combination of lower-exposure sectors that closely tracks ISIC4 2790 before the treatment year, then compares the post-treatment path of the treated sector to this synthetic counterfactual. Using both 2015 and 2016 as alternative treatment dates, the treated sector displays a post-treatment deviation consistent with lower markup growth relative to its synthetic control. The gap is largest shortly after the China shock and narrows toward the end of the sample, suggesting that the markup response is strongest in the short run and partly absorbed over time through firm adjustment, reallocation, or selective survival.

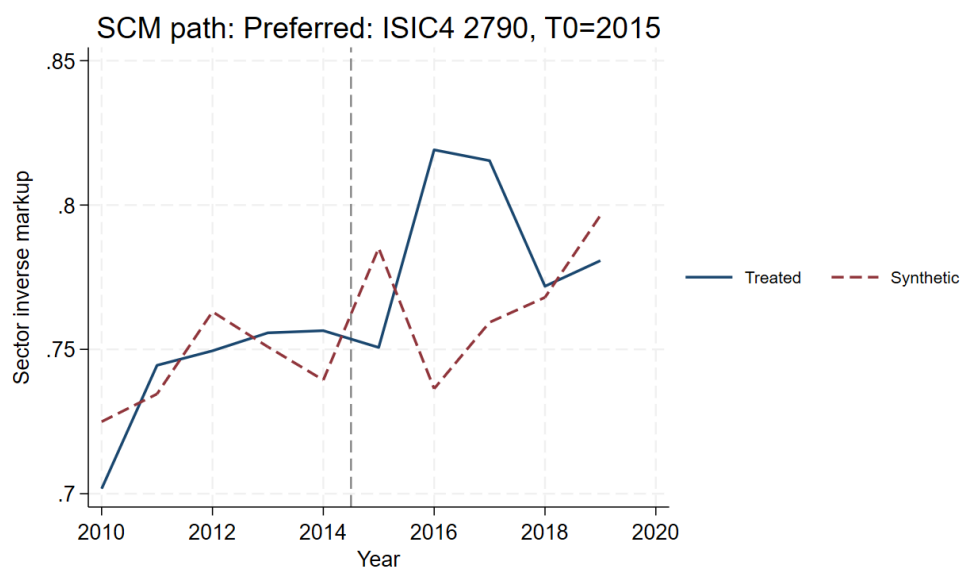


Figure 14: Synthetic-control path: ISIC4 2790, treatment year 2015

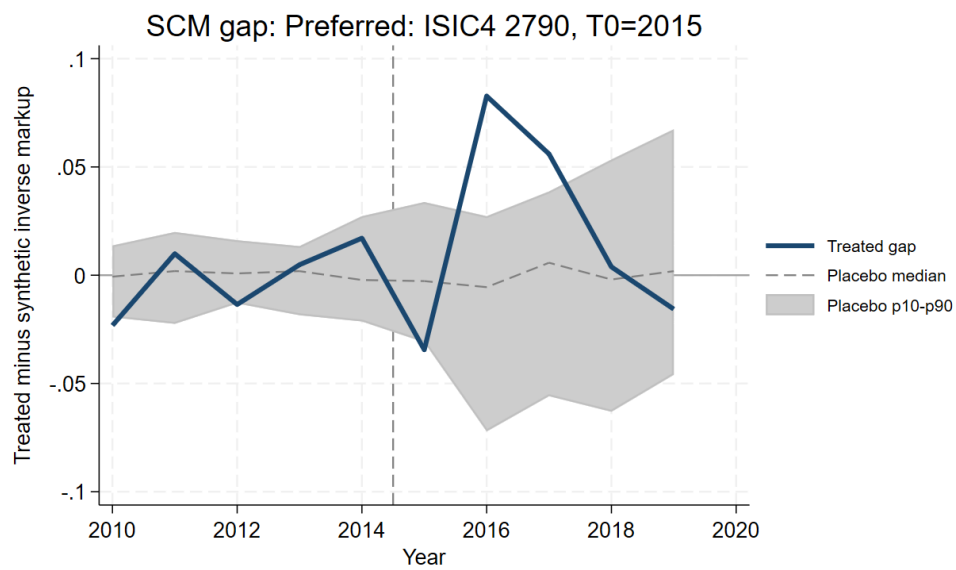


Figure 15: Synthetic-control gap: ISIC4 2790, treatment year 2015

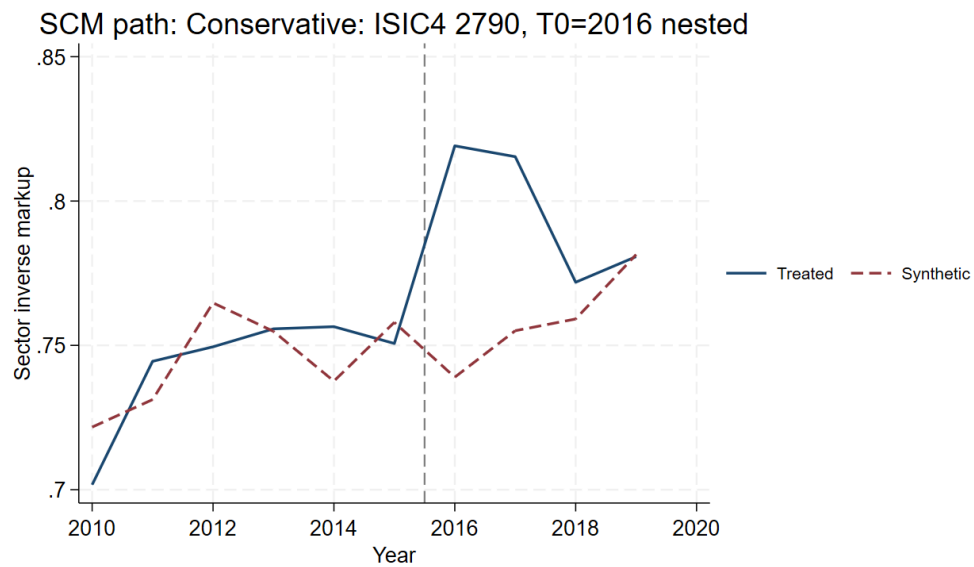


Figure 16: Synthetic-control path: ISIC4 2790, treatment year 2016

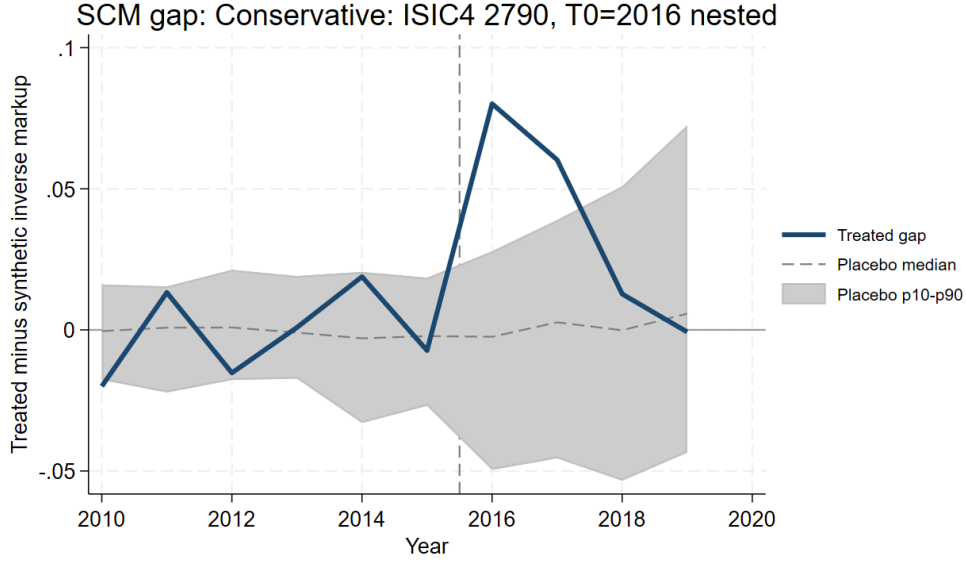


Figure 17: Synthetic-control gap: ISIC4 2790, treatment year 2016

B.3.11 Indirect-Inference Diagnostics

This subsection records the details behind Section 7.4. The main text reports the causal-core message: the model reproduces the output-competition IV response and passes a residual-IV validation. The appendix documents the estimator, the full parameter vector, the optimization path, non-targeted diagnostics, and the reason the current run should not be interpreted as a full welfare model.

The estimation is an indirect-inference exercise because the model is disciplined by auxiliary empirical objects rather than by a closed-form likelihood. For each candidate parameter vector θ , the model generates a simulated panel with the same observation structure as the data. The same auxiliary procedures are then applied to observed and simulated panels. In the notation of indirect inference, the binding function is

$$h(\theta) = \text{plim } m^s(\theta), \quad (26)$$

where $m^s(\theta)$ is the vector of simulated auxiliary moments. Identification comes from five output-competition moments: the baseline IV response of incumbent markup growth, grouped-IV effects at q80, q85, and q90, and the q90 interaction with sectoral concentration. The criterion is

$$Q(\theta) = [\hat{m} - m^s(\theta)]' W [\hat{m} - m^s(\theta)], \quad (27)$$

with W chosen to emphasize statistically stable causal moments. The fitted model is therefore

a model of the output-competition mechanism, not a model of every observed movement in markups. Sector inverse markups, decomposition objects, input-shock responses, selection diagnostics, and allocative-wedge objects are kept as diagnostics and appendix evidence, but they do not enter the criterion.

Table 30: Full parameter vector in the causal-core run

Parameter	Value	Status	Interpretation
output_scale	2.138	estimated	Scale of the output-competition channel
output_markup	-0.329	estimated	Baseline output-shock effect on markup growth
high_markup_output	-2.102	estimated	Extra output-shock effect for high-markup firms or sectors
concentration_output	-2.222	estimated	Extra output-shock effect in concentrated markets
share_output	-0.235	estimated	Output-shock component in market-share dynamics
share_input	0.000	fixed	Input channel switched off
input_markup	0.000	fixed	Input-markup channel switched off
mean_reversion	1.025	estimated	Auxiliary dynamic adjustment parameter
drift	0.350	estimated	Average residual drift
exit_intercept	-2.300	fixed	Baseline exit block fixed
exit_ip	0.000	fixed	No output-shock selection effect
exit_markup	0.000	fixed	No markup-dependent exit effect
eta	1.010	fixed	Elasticity parameter
nu	4.000	fixed	Elasticity parameter
rho	8.000	fixed	Elasticity parameter
lambda_min	0.050	fixed	Lower bound for lambda
lambda_max	0.980	fixed	Upper bound for lambda
default_lambda	0.850	fixed	Default lambda

The fixed zero input and selection parameters are important for interpretation. They make the exercise a clean test of the output-competition channel, while leaving input-shock and exit diagnostics for the next structural extension. Those non-targeted diagnostics do not overturn the targeted output-competition result.

The mean-reversion parameter is not a structural persistence estimate from the markup literature. It is an auxiliary dynamic adjustment parameter identified jointly from the targeted IV moments, so it should not be compared directly to AR(1) estimates of firm-level markup persistence.

The 2026-05-14 causal-core run re-estimates the five-moment criterion and reaches $Q = 1.099$. This improves on the earlier re-evaluation of the old parameter vector under the same five-moment criterion, and the fitted signs remain economically interpretable: output competition lowers markup growth, the effect is stronger where market power is initially high, and the residual-IV validation removes the output-shock signal from the fitted residual.

Table 31: Moment gaps entering the objective

Target moment	Data	Model	Gap	Weight
baseline_iv_dln_mu	-0.429	-0.414	-0.015	50.832
grouped_iv_q80	-0.803	-0.924	0.121	31.180
grouped_iv_q85	-1.236	-1.009	-0.227	11.768
grouped_iv_q90	-1.625	-1.572	-0.053	7.608
grouped_iv_q90_cr4_interaction	-5.371	-5.232	-0.139	0.369

The upper-tail pattern is useful diagnostically. The model gets the signs and broad magnitudes right, with an especially close fit at q90 and in the concentration-gradient moment. I do not add spline or tail-rank functions in this version, because the purpose of the exercise is to validate the narrow output-competition mechanism rather than to keep expanding the state-dependence block.

Table 32: Residual-IV validation

Validation regression	Coef.	SE	Obs.	Clusters	First-stage F
Residual markup growth on output shock	-0.052	0.295	31,375	90	22.930

Table 32 supports the interpretation that the model absorbs the output-shock variation it was designed to explain. The validation is targeted: it validates the fitted output-competition mechanism and leaves input, selection, and decomposition blocks for richer extensions.

The sector inverse-markup response is reported as an aggregation diagnostic rather than a targeted moment. The empirical estimate is less precise and is not consistently significant across the causal specifications, so it should not discipline the causal-core parameter vector. In the

diagnostic comparison, the data coefficient is 0.128 and the model coefficient is 0.298, implying that the model carries too much of the incumbent mechanism into the sales-weighted sector object.

Table 33: Selected non-targeted diagnostic moments

Diagnostic moment	Data	Model	Gap	Interpretation
Sector inverse-markup IV	0.128	0.298	-0.170	Secondary aggregation diagnostic
Input-shock between reallocation	8.055	2.429	5.625	Underpredicts input-shock reallocation
Input-shock within markup growth	0.142	0.017	0.126	Underpredicts input-markup relation
Exit rate	0.069	0.091	-0.022	Model exit rate too high
Exit IV	0.960	0.324	0.636	Underpredicts output-shock selection response

Table 33 clarifies the scope of the model. The data contain an input-shock reallocation pattern and an exit response that are outside the current causal-core block. Since the input and selection channels are switched off, these gaps are informative rather than damaging: they point directly to the next structural extension. The decomposition moments are also accounting diagnostics rather than targeted moments: within, reallocation, entry, and exit terms help describe changes in the sales-weighted inverse-markup object, but their relative shares can move with accounting definitions, sample windows, and observed-entry/exit conventions. In particular, small sales-weighted inverse-markup contributions of entry and exit do not imply that firm-count turnover is economically irrelevant. The model should not yet be used for welfare analysis, long-run industry composition, or full manufacturing-wide counterfactuals.

The no-output counterfactual removes the fitted output-competition channel. At the aggregate manufacturing level, the differences are small: the average inverse markup is 0.5775 in the observed model path and 0.5758 in the no-output counterfactual; the average aggregate markup is 1.7331 versus 1.7380. The 2011–2019 change in inverse markups is also close: -0.0191 in the observed path and -0.0196 in the no-output counterfactual. These numbers are diagnostics, not headline results.

The muted aggregate effect masks large sector-year effects. The largest differences in inverse markups occur in concentrated sector-years such as ISIC4 2790 in 2013, with a difference of 0.355, ISIC4 2513 in 2016, with a difference of 0.318, and ISIC4 1629 in 2013, with a difference of 0.236.

Other large sector-year differences change sign, so they offset in the manufacturing aggregate. The counterfactual therefore reinforces the thesis’s local interpretation: output competition disciplines exposed incumbent markups in particular market structures, but the current model does not say that Chinese competition explains the aggregate manufacturing markup path.

The detailed decomposition counterfactuals reinforce the same caution. In the non-targeted diagnostic moments, the model assigns absolute component shares of 0.150 to within-firm changes, 0.219 to between-firm reallocation, 0.318 to entry, and 0.312 to exit, while the data assign almost all absolute mass to within and between components. The no-output counterfactual gives very similar component shares to the observed model path. The correct conclusion is narrow and affirmative: the exercise validates the output-competition markup-discipline channel, while input, selection, decomposition, and welfare claims require a richer model.

B.4 Mechanism Evidence and Auxiliary Outcomes

B.4.1 State dependence and residual-demand evidence

Table 34 tests whether the output-competition effect varies with a firm state related to residual demand. The interaction is statistically meaningful, and the implied marginal effects remain negative at representative points of the lagged-markup distribution. This supports state dependence while leaving the stronger upper-tail interpretation to the grouped-IV quantile results in the main text.

B.4.2 Input-cost relief and reallocation among continuing firms

Table 35 tests whether sectors with stronger pre-period Chinese-input dependence respond differently to the input-supply and output-competition shocks. The input interaction is positive in the markup regression, consistent with upstream cost relief partly cushioning firms, while the output interaction is imprecise and weakly identified.

Table 36 tests whether output competition reallocates market share toward large incumbents, which would weaken the pro-competitive interpretation of the firm-level markup result. The output-shock interactions are not significant, while input-shock interactions are negative for large-firm and concentration proxies. This supports the main text’s claim that input exposure works through reallocation among continuing firms rather than through strengthened dominant survivors.

Table 34: Interaction with lagged firm markup

	(1)	Marginal effect of import competition		
	$\Delta \log \mu$	p25 μ_{t-1}	p50 μ_{t-1}	p75 μ_{t-1}
Panel A: IV coefficients				
Import penetration change (ΔIP)	-1.185** (0.452)			
$\Delta IP \times$ lagged log markup	11.52* (6.699)			
Input-supply shock (Z^{Input})	0.265 (0.592)			
Panel B: Implied marginal effects				
$\partial \Delta \log \mu / \partial \Delta IP$		-2.354	-1.597	-0.556
Observations	30551			
F-Stat	9.560			

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 35: Imported-input dependence heterogeneity

	(1)	(2)	(3)	(4)	(5)	(6)
	Markup Changes	Markup Changes	Revenue Changes	Revenue Changes	Sales Changes	Sales Changes
Output penetration	-0.454** (0.190)	-1.216* (0.681)	-1.632*** (0.535)	-1.974 (2.056)	-1.629** (0.631)	-3.227 (3.209)
Input supply	-2.041* (1.098)	0.285 (0.412)	0.996 (3.526)	2.651 (1.598)	-1.721 (3.388)	0.325 (1.977)
Input supply $\times$ China dependence	125.8** (53.42)		87.06 (205.2)		124.7 (159.2)	
Output penetration $\times$ China dependence		16.84 (11.23)		7.252 (33.32)		35.52 (56.91)
Observations	30551	30551	24345	24345	30551	30551
F-stat	27.71	2.743	32.03	2.845	27.71	2.743

Notes: Standard errors in parentheses. KP rk Wald F denotes the Kleibergen–Paap weak-instrument statistic for the excluded instrument(s). The output-penetration interaction columns have weak first stages and are interpreted as diagnostics only. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 36: Market-share reallocation toward large incumbents: output competition and input supply

	(1)	(2)	(3)	(4)	(5)	(6)
	$\Delta \log$	$\Delta \log$	$\Delta \log$	$\Delta \log$	$\Delta \log$	$\Delta \log$
	market share	market share	market share	market share	market share	market share
Output shock	0.0732					
× lagged firm size	(0.0853)					
Input shock		-0.843*				
× lagged firm size		(0.451)				
Output shock			0.803			
× large-firm indicator			(0.868)			
Input shock				-8.689**		
× large-firm indicator				(3.788)		
Output shock					1.787	
× CR4 membership					(1.379)	
Input shock						-12.43**
× CR4 membership						(6.086)
Observations	28159	35379	28159	35379	28159	35379
F-stat	33.40		0.904		3.238	

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

B.4.3 Auxiliary outcomes

Table 37 applies the mechanism design to revenue, domestic sales, and exports. The import-penetration coefficients are negative for revenue and domestic sales and imprecise for exports. These outcomes support the domestic-demand interpretation of the markup results and help rule out a story driven mainly by export redirection.

Table 37: Auxiliary outcomes: revenue, domestic sales, and exports

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	$\Delta \log \mu$	$\Delta \log \mu$	$\Delta \log$ Revenue	$\Delta \log$ Revenue	$\Delta \log$ Domestic sales	$\Delta \log$ Domestic sales	$\Delta \log$ Exports	$\Delta \log$ Exports
Import penetration change (ΔIP)	-0.540** (0.231)	-0.979 (0.927)	-1.936*** (0.637)	-1.198 (3.075)	-1.627** (0.673)	-1.515 (3.902)	-0.416 (2.625)	-11.75 (17.31)
Input-supply shock (Z^{Input})	-6.864 (5.884)	0.373 (0.429)	-25.19* (14.52)	2.866* (1.661)	1.743 (18.51)	0.847 (2.009)	53.26 (35.05)	-10.10* (5.985)
Input-supply shock $\times$ imported-input dependence	28.65 (23.00)		107.9* (57.56)		-3.609 (70.00)		-234.2* (136.7)	
Import penetration $\times$ imported-input dependence		1.861 (2.849)		-1.554 (9.161)		-0.438 (12.14)		36.71 (52.04)
Observations	30551	30551	24345	24345	30551	30551	19327	19327
KP rk Wald F	—	—	—	—	—	—	—	—

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$